

Answers to Questions from the RMS

Contents

1	Question Q587	5
1.1	Audited informal answer	5
1.1.1	1. Faithful restatement and conventions	5
1.1.2	2. Divided differences	6
1.2	3. Part <i>a</i>	8
1.2.1	Theorem 3.1	8
1.3	4. Part <i>b</i>	9
1.3.1	Theorem 4.1	9
1.4	5. Differentiating a DIT	11
1.4.1	Lemma 5.1 — Differentiation theorem	11
1.5	6. Part <i>c</i>	14
1.6	7. Part <i>d</i>	14
1.6.1	Lemma 7.1 — DIT(1) everywhere implies C^1	14
1.6.2	Proposition 7.2	15
1.6.3	8. The order-zero ambiguity	16
1.7	9. Checks and sharpness	17
1.8	Final classification	19
2	Question Q604	20
2.1	Audited informal answer	20
2.1.1	1. Faithful restatement	20
2.1.2	2. Explicit answer	21
2.1.3	3. The universal binomial Bézout identity	22
2.1.4	4. Proof of the formula	23
2.1.5	5. Uniqueness and classification of all pairs	24
2.1.6	6. Symmetric remainder formula	25
2.1.7	7. A directly degree-normalized B -adic formula	26
2.1.8	8. Small and boundary cases	27
2.1.9	9. Why the Euclidean reduction is genuinely necessary	28
2.1.10	10. Hypothesis audit	29

3	Question Q701	30
3.1	Audited informal answer	30
3.1.1	1. Faithful restatement	30
3.1.2	2. The slack pseudometric	30
3.1.3	3. Complete characterization	31
3.1.4	4. Proof of the characterization	32
3.1.5	5. Quotient-space form of the answer	35
3.1.6	6. Saturated pairs and a simple sufficient condition . .	35
3.1.7	7. The finite-set classification	36
3.1.8	8. Explicit extreme points	37
3.1.9	9. McShane-type construction of further extreme points	38
3.1.10	10. Necessary conditions, and why they are not sufficient	39
3.1.11	11. Strong extremality	40
3.1.12	12. Boundary cases and necessity of the hypotheses . .	41
3.1.13	13. Compactness, separation, countability, and choice audit	42
3.1.14	Final classification	43
4	Question Q668	43
4.1	Audited informal answer	44
4.2	Partial result	44
4.2.1	1. Faithful restatement	44
4.3	2. Main classification theorem	45
4.3.1	Theorem 2.1	46
4.4	3. Proof of Theorem 2.1	47
4.4.1	3.1. Monotonicity	47
4.4.2	3.2. Strict decrease unless all entries are extremal . .	47
4.4.3	3.3. Every limit point is binary up to scale	49
4.4.4	3.4. Algebra over $\mathbb{F}_2$	49
4.4.5	3.5. Why limit points belong to $\mathcal{S}_n$	50
4.4.6	3.6. P is a permutation of $\mathcal{S}_n$	51
4.4.7	3.7. The orbit shadows one periodic cycle	52
4.4.8	3.8. All real periodic points	53
4.5	4. Complete asymptotic description for $n = 5$	53
4.5.1	Proposition 4.1	54
4.5.2	Corollary 4.2	55
4.6	5. A nonconstant 5-tuple converging to zero	56
4.7	6. An exact non-eventually-periodic $n = 5$ orbit with nonzero attractor	57
4.7.1	Theorem 6.1	57

4.7.2	Exact finite verifier	60
4.8	7. Exact algorithm for rational starting vectors	64
4.8.1	Consequence for rational 5-tuples	65
4.9	8. Small and boundary cases	66
4.10	9. Exact remaining obstruction	67
5	Question Q728	68
5.1	Audited informal answer	68
5.1.1	1. Faithful restatement and a domain issue	68
5.1.2	2. Elementary terminal identities	69
5.2	3. The control interval of a position	70
5.2.1	4. Effect of one move	70
5.3	5. The steering lemma	72
5.4	6. The interval-control theorem	74
5.5	7. Control intervals for two initial pieces	75
5.6	8. Determination of the winner	76
5.6.1	Case 1: $n = 3q + 1$	76
5.6.2	Case 2: $n = 3q + 2$	77
5.6.3	Case 3: $n = 3q$	78
5.7	9. Explicit optimal strategy and complexity	79
5.8	10. Boundary checks	81
5.9	11. Necessity of the hypotheses	81
5.10	12. Exact computational verification	82
5.10.1	Final conclusion	83
6	Question Q756	84
6.1	Audited informal answer	84
6.1.1	1. Faithful restatement	84
6.1.2	2. Final theorem	84
6.2	3. Automatic regularity	85
6.3	4. Polynomial solutions	86
6.4	5. Extension from a fundamental annulus	87
6.4.1	5.1. Extension toward infinity	87
6.4.2	5.2. Extension toward the origin	88
6.5	6. Uniform boundedness near the origin	89
6.6	7. Killing both one-sided limits	90
6.6.1	Infinite-dimensionality	91
6.7	8. Analytic solutions	91
6.8	9. Boundary and small-case checks	93
6.8.1	Conclusion	94

7	Question Q759	94
7.1	Audited informal answer	94
7.1.1	Faithful restatement	94
7.1.2	Final result	95
7.2	1. Reduction to functions on $([0,1])$	95
7.2.1	1.1. Smoothness and flatness of β	96
7.3	2. The first sequence	97
7.3.1	2.1. Elementary derivative bounds	97
7.3.2	2.2. Boundary layer	98
7.3.3	2.3. Interior region	99
7.4	3. A weighted binomial approximation	100
7.4.1	Lemma 3.1	100
7.5	4. Derivatives of the geometric partial sums	102
7.5.1	Lemma 4.1	102
7.6	5. The second sequence	103
7.7	6. Boundary checks, small cases, and sharpness	105
7.7.1	6.1. At $x = 0$	105
7.7.2	6.2. At $x = \pm 1$	105
7.7.3	6.3. Sharpness for part b	106
7.7.4	6.4. Role of the domain	106
7.8	Conclusion	107
8	Question Q764	107
8.1	Audited informal answer	107
8.1.1	1. Faithful restatement	107
8.1.2	2. Final result	108
8.2	Part a : points on the real line	109
8.2.1	3. The optimal cost of one consecutive block	109
8.2.2	4. Reduction to consecutive blocks	109
8.2.3	5. Dynamic programming recurrence	110
8.2.4	6. Monotonicity of the transition	111
8.2.5	7. Precomputing all block costs in $O(n^2)$	113
8.2.6	8. Exact quadratic-time algorithm	113
8.2.7	9. A simpler independent decision algorithm	116
8.3	Part b : an arbitrary finite metric space	116
8.3.1	10. Threshold-graph characterization	117
8.3.2	11. An exact finite algorithm	117
8.3.3	12. NP-completeness of the general metric problem	118
8.3.4	13. A polynomial-time factor-2 algorithm	120
8.3.5	14. The factor 2 cannot be improved in polynomial time	121

8.4	15. Boundary and sanity checks	122
8.5	Final answer	123

1 Question Q587

Informal verdict. **A — Complete solution**, with **high confidence** for every substantive case $n \geq 1$ and $1 \leq p < n$.

Lean coverage. The Lean development proves parts *a–d*, the Peano expansion, the differentiation identities, and the order-zero counterexample. It models the interval as the whole real line and documents the resulting endpoint and node-ordering conventions.

1.1 Audited informal answer

1.1.1 1. Faithful restatement and conventions

Let $I \subset \mathbb{R}$ be an interval and $f : I \rightarrow \mathbb{R}$. For distinct nodes

$$x_0 < x_1 < \cdots < x_n$$

in I , let $P_{x_0, \dots, x_n}$ be the Lagrange interpolation polynomial of f at those nodes, written in the ordinary monomial basis as

$$P_{x_0, \dots, x_n}(X) = \sum_{q=0}^n c_q(x_0, \dots, x_n) X^q.$$

The function f is said to admit an interpolated Taylor expansion of order n , abbreviated $\text{DIT}(n)$, at $x \in I$, when every $c_q(x_0, \dots, x_n)$ has a finite limit as all nodes tend to x . The notation $f^{[q]}(x)$ denotes $(q!)$ times that limit. The four questions are:

1. Show that it is enough to assume convergence of the highest coefficient c_n .
2. Show that $\text{DIT}(n)$ implies an ordinary asymptotic expansion of order n .
3. If $p < n$ and $f \in C^p$, prove that f has $\text{DIT}(n)$ at x if and only if $f^{(p)}$ has $\text{DIT}(n - p)$ there.
4. If f has $\text{DIT}(p)$ at every point, determine whether $\text{DIT}(n)$ for f is equivalent to $\text{DIT}(n - p)$ for the resulting function $f^{[p]}$.

There are three minor conventions to make explicit.

- “Increasing order” must mean **strictly** increasing, since ordinary Lagrange interpolation requires distinct nodes.
- Limits at an endpoint of I are relative, hence one-sided. Likewise, $C^p(I)$ means C^p on the interior, with the corresponding one-sided continuous extensions at finite endpoints.
- Because the polynomial is expanded in powers of X , rather than $X - x$, the lower quantities $f^{[q]}(x)$, $q < n$, depend on the ambient order n . When necessary, they will be denoted $f_{(n)}^{[q]}(x)$. The top quantity $f^{[n]}(x)$ is intrinsic.

There is also a genuine order-zero convention issue, addressed at the end.

1.1.2 2. Divided differences

For pairwise distinct $t_0, \dots, t_r \in I$, write

$$[t_0, \dots, t_r]f := \sum_{i=0}^r \frac{f(t_i)}{\prod_{\substack{0 \leq j \leq r \\ j \neq i}} (t_i - t_j)}.$$

This is the divided difference of order r . It is symmetric in its arguments.

From the Lagrange formula,

$$P_{t_0, \dots, t_r}(X) = \sum_{i=0}^r f(t_i) \prod_{j \neq i} \frac{X - t_j}{t_i - t_j},$$

the leading coefficient is

$$c_r(t_0, \dots, t_r) = [t_0, \dots, t_r]f.$$

We shall repeatedly use the replacement identity

$$[A, u]f - [A, v]f = (u - v)[A, u, v]f, \quad (2.1)$$

where A is any list of distinct common nodes and all displayed nodes are pairwise distinct. It follows immediately from the usual recursion for divided differences, or from the explicit formula above.

Lemma 2.1 — Descent of divided differences Let $r \geq 1$, and suppose that the order- r divided differences of f are bounded near x : there exist $M, \rho_0 > 0$ such that

$$|[t_0, \dots, t_r]f| \leq M$$

whenever the nodes are distinct and belong to

$$U_{\rho_0} := I \cap (x - \rho_0, x + \rho_0).$$

Then the order- $(r-1)$ divided differences have a finite limit as their r nodes tend to x .

Proof Fix $0 < \rho < \rho_0$, and let

$$A = (a_0, \dots, a_{r-1}), \quad B = (b_0, \dots, b_{r-1})$$

be two r -tuples of distinct nodes in U_ρ .

Choose distinct auxiliary nodes

$$C = (c_0, \dots, c_{r-1}) \subset U_\rho$$

disjoint from all the nodes occurring in A and B . This is possible because a nondegenerate real interval contains infinitely many points in every relative neighborhood of x .

Replace the nodes of A one at a time by the corresponding nodes of C . At each step, (2.1) gives

$$\left| [A^{(j+1)}]f - [A^{(j)}]f \right| \leq |c_j - a_j|M \leq 2\rho M.$$

There are r replacements, hence

$$|[A]f - [C]f| \leq 2r\rho M.$$

Similarly,

$$|[B]f - [C]f| \leq 2r\rho M.$$

Therefore

$$|[A]f - [B]f| \leq 4r\rho M. \tag{2.2}$$

Thus the oscillation of the order- $(r-1)$ divided difference over all r -tuples in U_ρ tends to zero with ρ . The Cauchy criterion gives a unique finite limit as all nodes tend to x . $\square$

Consequently, if the order- n divided difference has a finite limit at x , then successively all lower divided differences do. We write

$$\lambda_k(f; x) := \lim_{t_0, \dots, t_k \rightarrow x} [t_0, \dots, t_k]f, \quad 0 \leq k \leq n. \quad (2.3)$$

For $n \geq 1$, the limit $\lambda_0(f; x)$ equals $f(x)$. Indeed,

$$f(t) - f(x) = (t - x)[x, t]f,$$

and the first divided differences remain bounded near x .

1.2 3. Part *a*

1.2.1 Theorem 3.1

For $n \geq 1$, the following are equivalent:

1. f has DIT(n) at x ;
2. the highest coefficient $c_n(x_0, \dots, x_n)$ has a finite limit as all nodes tend to x ;
3. the order- n divided difference

$$[x_0, \dots, x_n]f$$

has a finite limit as all nodes tend to x .

Moreover, if the limits $\lambda_k(f; x)$ are defined by (2.3), the interpolation polynomials converge coefficientwise to

$$Q_{n,x}(X) := \sum_{k=0}^n \lambda_k(f; x)(X - x)^k. \quad (3.1)$$

Proof The equivalence of 2 and 3 follows from

$$c_n(x_0, \dots, x_n) = [x_0, \dots, x_n]f.$$

Assume that this highest coefficient has a finite limit. It is then bounded near the diagonal. Lemma 2.1, applied successively, shows that all limits

$$\lambda_k(f; x), \quad k = 0, \dots, n,$$

exist.

For any ordering of the distinct nodes, Newton's interpolation formula is

$$P_{x_0, \dots, x_n}(X) = \sum_{k=0}^n [x_0, \dots, x_k] f \prod_{j=0}^{k-1} (X - x_j). \quad (3.2)$$

As all nodes tend to x ,

$$[x_0, \dots, x_k] f \longrightarrow \lambda_k(f; x)$$

and

$$\prod_{j=0}^{k-1} (X - x_j) \longrightarrow (X - x)^k$$

coefficientwise. Hence (3.2) converges coefficientwise to $Q_{n,x}$. In particular, all ordinary monomial coefficients c_q have finite limits.

The converse is immediate because $\text{DIT}(n)$ includes convergence of c_n . $\square$

Expanding (3.1) in powers of X , we obtain the explicit answer

$$\lim c_q(x_0, \dots, x_n) = \sum_{k=q}^n \binom{k}{q} (-x)^{k-q} \lambda_k(f; x). \quad (3.3)$$

Thus

$$f_{(n)}^{[q]}(x) = q! \sum_{k=q}^n \binom{k}{q} (-x)^{k-q} \lambda_k(f; x). \quad (3.4)$$

In particular,

$$f^{[n]}(x) = n! \lambda_n(f; x). \quad (3.5)$$

Formula (3.4) also explains why the lower $f_{(n)}^{[q]}$ generally depend on n .

1.3 4. Part b

1.3.1 Theorem 4.1

If f has $\text{DIT}(n)$ at x , with $n \geq 1$, then

$$f(x+h) = \sum_{k=0}^n \lambda_k(f; x) h^k + o(|h|^n) \quad (h \rightarrow 0, x+h \in I). \quad (4.1)$$

Equivalently,

$$f(t) = Q_{n,x}(t) + o(|t-x|^n), \quad (4.2)$$

where $Q_{n,x}$ is the coefficientwise limit of the interpolation polynomials.

Proof Fix $h \neq 0$, with $x + h \in I$. Choose n pairwise distinct auxiliary nodes

$$u_0^{(m)}, \dots, u_{n-1}^{(m)} \in I \setminus \{x + h\}$$

such that

$$u_j^{(m)} \longrightarrow x \quad (m \rightarrow \infty)$$

for every j .

Apply Newton's formula to the nodes

$$u_0^{(m)}, \dots, u_{n-1}^{(m)}, x + h$$

and evaluate at $X = x + h$:

$$\begin{aligned} f(x + h) &= \sum_{k=0}^{n-1} [u_0^{(m)}, \dots, u_k^{(m)}] f \prod_{j=0}^{k-1} (x + h - u_j^{(m)}) \\ &\quad + [u_0^{(m)}, \dots, u_{n-1}^{(m)}, x + h] f \prod_{j=0}^{n-1} (x + h - u_j^{(m)}). \end{aligned} \quad (4.3)$$

As $m \rightarrow \infty$, the terms of order $k < n$ tend to

$$\lambda_k(f; x) h^k,$$

and the final product tends to $h^n \neq 0$. Hence the last divided difference in (4.3) has a limit, namely

$$R(h) := \frac{f(x + h) - \sum_{k=0}^{n-1} \lambda_k(f; x) h^k}{h^n}. \quad (4.4)$$

We claim that

$$R(h) \longrightarrow \lambda_n(f; x).$$

Given $\varepsilon > 0$, choose $\delta > 0$ such that

$$|[t_0, \dots, t_n]f - \lambda_n(f; x)| < \varepsilon \quad (4.5)$$

whenever all the distinct t_i lie in $I \cap (x - \delta, x + \delta)$.

If $0 < |h| < \delta$, then, for sufficiently large m , all the auxiliary nodes in (4.3), as well as $x + h$, lie in this neighborhood. Thus the last divided difference in (4.3) differs from $\lambda_n(f; x)$ by less than ε . Passing to its limit gives

$$|R(h) - \lambda_n(f; x)| \leq \varepsilon.$$

This proves (4.1). $\square$

In terms of the coefficients from the statement,

$$Q_{n,x}(X) = \sum_{q=0}^n \frac{f_{(n)}^{[q]}(x)}{q!} X^q,$$

so another exact formulation of the requested expansion is

$$f(t) = \sum_{q=0}^n \frac{f_{(n)}^{[q]}(x)}{q!} t^q + o(|t-x|^n). \quad (4.6)$$

The centered coefficients $\lambda_k(f; x)$ are the more intrinsic ones.

1.4 5. Differentiating a DIT

The central result for parts *c* and *d* is the following one-step equivalence.

1.4.1 Lemma 5.1 — Differentiation theorem

Let $r \geq 1$, and let f be C^1 on a relative neighborhood of x . Then

$$f \text{ has DIT}(r) \text{ at } x \iff f' \text{ has DIT}(r-1) \text{ at } x. \quad (5.1)$$

When these conditions hold,

$$\lambda_{r-1}(f'; x) = r, \lambda_r(f; x). \quad (5.2)$$

In fact, continuity of (f') is more than is needed; differentiability on a neighborhood suffices.

Proof: from f to (f') Assume

$$\lambda_r(f; x) = L.$$

Let

$$y_0 < \cdots < y_{r-1}$$

be distinct nodes near x , and define

$$G(t) := [y_0 + t, \dots, y_{r-1} + t]f.$$

Because all differences $(y_i + t) - (y_j + t)$ are independent of t , the explicit divided-difference formula gives

$$G'(0) = [y_0, \dots, y_{r-1}]f'. \quad (5.3)$$

For sufficiently small nonzero h , telescope by shifting the nodes one at a time:

$$G(h) - G(0) = \sum_{j=0}^{r-1} \left([y_0 + h, \dots, y_j + h, y_{j+1}, \dots, y_{r-1}]f \right. \\ \left. - [y_0 + h, \dots, y_{j-1} + h, y_j, \dots, y_{r-1}]f \right)$$

By the replacement identity (2.1), each summand equals

$$h[Z_j(h)]f,$$

where $Z_j(h)$ is the $(r+1)$ -tuple consisting of the common $r-1$ nodes together with y_j and $y_j + h$. Hence

$$\frac{G(h) - G(0)}{h} = \sum_{j=0}^{r-1} [Z_j(h)]f. \quad (5.4)$$

Let $\varepsilon > 0$. If all y_i are sufficiently close to x , then for all sufficiently small h , every divided difference on the right of (5.4) differs from L by at most ε/r . Therefore

$$\left| \frac{G(h) - G(0)}{h} - rL \right| < \varepsilon.$$

Letting $h \rightarrow 0$ and using (5.3),

$$|[y_0, \dots, y_{r-1}]f' - rL| \leq \varepsilon.$$

Thus

$$[y_0, \dots, y_{r-1}]f' \longrightarrow rL,$$

which, by part *a*, is precisely $\text{DIT}(r-1)$ for (f') .

Proof: from (f') to f Conversely, assume

$$\lambda_{r-1}(f'; x) = M.$$

Take arbitrary nodes

$$z_0 < z_1 < \dots < z_r$$

near x , and let P be the degree- $\leq r$ interpolation polynomial of f at these nodes. The function

$$H := f - P$$

vanishes at each z_i . By Rolle's theorem, for every $i = 0, \dots, r-1$, there is a point

$$\xi_i \in (z_i, z_{i+1})$$

such that

$$H'(\xi_i) = 0.$$

Thus

$$P'(\xi_i) = f'(\xi_i).$$

The points satisfy

$$\xi_0 < \dots < \xi_{r-1},$$

and (P') , of degree at most $r-1$, is therefore the interpolation polynomial of (f') at these r points.

The leading coefficient of (P') is

$$r[z_0, \dots, z_r]f,$$

while the leading coefficient of the interpolation polynomial of (f') is

$$[\xi_0, \dots, \xi_{r-1}]f'.$$

Consequently,

$$r[z_0, \dots, z_r]f = [\xi_0, \dots, \xi_{r-1}]f'. \quad (5.5)$$

As all $z_i \rightarrow x$, all $\xi_i \rightarrow x$. Therefore the right-hand side of (5.5) tends to M , and

$$[z_0, \dots, z_r]f \longrightarrow \frac{M}{r}.$$

Part *a* now gives $\text{DIT}(r)$ for f , with

$$\lambda_r(f; x) = \frac{M}{r}.$$

This completes the proof. $\square$

1.5 6. Part *c*

Let $0 \leq p < n$ and $f \in C^p(I)$.

For $p = 0$, the assertion is tautological. Suppose $p \geq 1$. Apply Lemma 5.1 successively to

$$f, f', f'', \dots, f^{(p-1)}$$

with orders

$$n, n-1, n-2, \dots, n-p+1.$$

We obtain

$$\begin{aligned} f \text{ has DIT}(n) \text{ at } x &\iff f' \text{ has DIT}(n-1) \text{ at } x \\ &\iff f'' \text{ has DIT}(n-2) \text{ at } x \\ &\vdots \\ &\iff f^{(p)} \text{ has DIT}(n-p) \text{ at } x. \end{aligned}$$

This proves the required equivalence.

The limiting highest coefficients satisfy the precise identity

$$\lambda_{n-p}(f^{(p)}; x) = n(n-1) \cdots (n-p+1) \lambda_n(f; x) \frac{n!}{(n-p)!} \lambda_n(f; x). \quad (6.1)$$

Multiplying by $(n-p)!$, we get the particularly simple relation

$$\boxed{(f^{(p)})^{[n-p]}(x) = f^{[n]}(x).} \quad (6.2)$$

Thus part *c* holds, and (6.1) gives the exact transformation of the top divided-difference limit.

1.6 7. Part *d*

The substantive answer is affirmative for every $1 \leq p < n$.

The point is that having DIT(p) at every point automatically forces f to be C^p , and $f^{[p]}$ is then exactly the ordinary derivative $f^{(p)}$.

1.6.1 Lemma 7.1 — DIT(1) everywhere implies C^1

Suppose g has DIT(1) at every point of I . Then $g \in C^1(I)$, and

$$g'(x) = \lambda_1(g; x) = g^{[1]}(x).$$

Proof At a fixed x ,

$$\frac{g(t) - g(x)}{t - x} = [x, t]g \longrightarrow \lambda_1(g; x),$$

so g is differentiable and

$$g'(x) = \lambda_1(g; x).$$

It remains to prove continuity of (g') . Given $\varepsilon > 0$, the strict secant limit at x provides $\delta > 0$ such that

$$|[u, v]g - g'(x)| < \varepsilon$$

whenever $u \neq v$ and both (u, v) lie within δ of x .

For y sufficiently close to x , let $v \rightarrow y$. All secants $([y, v]g)$ are then within ε of $(g'x)$. Passing to the limit $v \rightarrow y$ gives

$$|g'(y) - g'(x)| \leq \varepsilon.$$

Thus (g') is continuous. $\square$

1.6.2 Proposition 7.2

Let $p \geq 1$. If f has DIT(p) at every point of I , then

$$f \in C^p(I)$$

and

$$\boxed{f^{[p]} = f^{(p)}}. \tag{7.1}$$

Proof Since DIT(p) implies, by the descent argument in part *a*, existence of the first divided-difference limit, f has DIT(1) at every point. Lemma 7.1 therefore gives

$$f \in C^1(I).$$

By Lemma 5.1, because f has DIT(p) everywhere,

$$f'$$

has DIT($p - 1$) everywhere. If $p - 1 \geq 1$, Lemma 7.1 applied to (f') gives $f' \in C^1$, hence $f \in C^2$.

Continuing inductively,

$$f^{(j)}$$

has $\text{DIT}(p-j)$ everywhere and is C^1 for $j = 0, \dots, p-1$. Hence

$$f \in C^p(I).$$

The factor relation (5.2), iterated p times, gives

$$f^{(p)}(x) = p! \lambda_p(f; x).$$

Since c_p is the order- p divided difference,

$$f^{[p]}(x) = p! \lambda_p(f; x).$$

Therefore

$$f^{[p]}(x) = f^{(p)}(x)$$

at every point. $\square$

We may now apply part *c*:

$$\begin{aligned} f \text{ has } \text{DIT}(n) \text{ at } x &\iff f^{(p)} \text{ has } \text{DIT}(n-p) \text{ at } x \\ &\iff f^{[p]} \text{ has } \text{DIT}(n-p) \text{ at } x. \end{aligned}$$

Hence:

$$\boxed{\text{For every } 1 \leq p < n, \text{ the answer to part (d) is yes.}} \quad (7.2)$$

The global assumption “at every point of I ” can be weakened: for the equivalence at a fixed x , it is enough that f have $\text{DIT}(p)$ throughout a relative neighborhood of x .

1.6.3 8. The order-zero ambiguity

If $p = 0$, the answer depends on what “ $\text{DIT}(0)$ ” means.

For $n = 0$, the interpolation polynomial is simply

$$P_{x_0}(X) = f(x_0).$$

Thus the literal condition in the PDF is merely the existence of

$$\lim_{t \rightarrow x} f(t).$$

Continuity convention If $\text{DIT}(0)$ is understood to require

$$\lim_{t \rightarrow x} f(t) = f(x),$$

then

$$f^{[0]} = f,$$

and part d for $p = 0$ is tautologically true.

Punctured-limit convention If only the finite punctured limit is required, part d is false for $p = 0$.

Take

$$I = (-1, 1), \quad f(t) = \begin{cases} 1, & t = 0, \\ 0, & t \neq 0. \end{cases}$$

At every point, the punctured limit exists, and

$$f^{[0]} \equiv 0.$$

Thus $f^{[0]}$ has $\text{DIT}(1)$ at 0. But f does not, since for $t > 0$,

$$[0, t]f = \frac{f(t) - f(0)}{t} = -\frac{1}{t},$$

which diverges.

Therefore the fully literal conclusion is:

Part (d) is affirmative for every $p \geq 1$. [2mm] For $p = 0$, it is affirmative under the continuity convention but false under the punctured-limit convention.

The same convention affects part b when $n = 0$: a punctured finite limit gives an expansion with a constant equal to that limit, but not necessarily with constant $f(x)$.

1.7 9. Checks and sharpness

9.1 The case $n = 1$ $\text{DIT}(1)$ at x means

$$[u, v]f = \frac{f(v) - f(u)}{v - u}$$

has a limit as $u, v \rightarrow x$. This is strict differentiability at x . The results give

$$f(x + h) = f(x) + \lambda_1(f; x)h + o(|h|),$$

and, if this condition holds everywhere, $f \in C^1$.

9.2 Polynomial check For a polynomial f , all the limits exist. The limiting interpolation polynomial is its Taylor polynomial of degree n at x :

$$Q_{n,x}(X) = \sum_{k=0}^n \frac{f^{(k)}(x)}{k!} (X - x)^k,$$

with the sum truncated automatically if $\deg f < n$. Thus

$$\lambda_k(f; x) = \frac{f^{(k)}(x)}{k!},$$

and (6.1) becomes

$$\frac{f^{(n)}(x)}{(n-p)!} = \frac{n!}{(n-p)!} \frac{f^{(n)}(x)}{n!},$$

as required.

9.3 The converse to part b is false An ordinary asymptotic expansion need not imply a DIT.

Let

$$g(0) = 0, \quad g(t) = t^2 \sin(1/t) \quad (t \neq 0).$$

Then

$$g(t) = o(t),$$

so g has a first-order expansion at 0 with derivative 0. But

$$g'(t) = 2t \sin(1/t) - \cos(1/t)$$

oscillates between values close to 1 and (-1) as $t \rightarrow 0$.

At points $t_k \rightarrow 0$ where $g'(t_k) \rightarrow 1$, choose $h_k \rightarrow 0$ so that

$$[t_k, t_k + h_k]g \rightarrow 1.$$

At another sequence $s_k \rightarrow 0$, choose secants tending to (-1). Hence the two-point divided differences do not have a unique limit as both nodes tend to 0. Thus g does not have DIT(1) there.

So DIT is strictly stronger than the existence of an ordinary Peano–Taylor expansion.

9.4 Role of the hypotheses

- Distinctness of the nodes is indispensable for ordinary Lagrange interpolation. Repeated nodes would require Hermite interpolation.
 - The interval assumption is used essentially in the Rolle-theorem argument: each consecutive pair of interpolation nodes determines an intervening point.
 - The C^p assumption in part *c* is sufficient but not optimal. The proof only needs p -fold differentiability on a neighborhood of x .
 - In part *d*, the “every point” hypothesis is what upgrades the strict divided-difference regularity to C^p . Locally around x is sufficient.
 - Endpoint cases require no separate argument beyond taking relative limits and one-sided derivatives.
-

1.8 Final classification

For $n \geq 1$, let

$$\Lambda_n(f; x) = \lim_{x_0, \dots, x_n \rightarrow x} [x_0, \dots, x_n]f.$$

Then:

$$\boxed{f \text{ has DIT}(n) \text{ at } x \iff \Lambda_n(f; x) \in \mathbb{R}.}$$

When this holds, all lower limits $\lambda_k(f; x)$ exist, the interpolation polynomials converge to

$$Q_{n,x}(X) = \sum_{k=0}^n \lambda_k(f; x)(X - x)^k,$$

and

$$\boxed{f(x + h) = \sum_{k=0}^n \lambda_k(f; x)h^k + o(|h|^n).}$$

For $0 \leq p < n$ and $f \in C^p$,

$$\boxed{f \text{ has DIT}(n) \text{ at } x \iff f^{(p)} \text{ has DIT}(n - p) \text{ at } x,}$$

with

$$\lambda_{n-p}(f^{(p)}; x) = \frac{n!}{(n-p)!} \lambda_n(f; x).$$

Finally, if $p \geq 1$ and f has $\text{DIT}(p)$ at every point, then

$$f \in C^p, \quad f^{[p]} = f^{(p)},$$

and therefore

$$f \text{ has } \text{DIT}(n) \text{ at } x \iff f^{[p]} \text{ has } \text{DIT}(n-p) \text{ at } x.$$

Formal verification. Lean source; online type-check.

2 Question Q604

Informal verdict. A — **Complete solution**, with **very high confidence**.

Lean coverage. The Lean development proves the printed normalized Bezout-pair theorem, its uniqueness, the universal binomial identity, the classification of all pairs, the worked example, and the constant-polynomial boundary case over an arbitrary field.

2.1 Audited informal answer

2.1.1 1. Faithful restatement

Let $\mathbb{K}$ be a field. Let $A, B \in \mathbb{K}[X]$ be coprime, and let $U, V \in \mathbb{K}[X]$ satisfy

$$\deg U < \deg B, \quad AU + BV = 1.$$

For positive integers (m, n) , determine explicitly the pair $(U_{m,n}, V_{m,n}) \in \mathbb{K}[X]^2$ such that

$$\deg U_{m,n} < \deg B^n \quad \text{and} \quad A^m U_{m,n} + B^n V_{m,n} = 1.$$

This is exactly Q604 on the rendered PDF page. The notation d° denotes polynomial degree. There is no apparent typo. The statement leaves implicit the standard convention that $\mathbb{K}$ is a field; this is the setting used below. We use $\deg 0 = -\infty$.

2.1.2 2. Explicit answer

For integers $r \geq 1$ and $s \geq 1$, define

$$S_{r,s}(T) := \sum_{j=0}^{s-1} \binom{r+j-1}{j} T^j \in \mathbb{K}[T],$$

where the integer binomial coefficients are mapped into $\mathbb{K}$ through the canonical homomorphism $\mathbb{Z} \rightarrow \mathbb{K}$.

Set

$$\widehat{U}_{m,n} := U^m S_{m,n}(BV) = U^m \sum_{j=0}^{n-1} \binom{m+j-1}{j} (BV)^j$$

and

$$\widehat{V}_{m,n} := V^n S_{n,m}(AU) = V^n \sum_{i=0}^{m-1} \binom{n+i-1}{i} (AU)^i.$$

Perform the Euclidean division

$$\widehat{U}_{m,n} = B^n Q_{m,n} + R_{m,n}, \quad \deg R_{m,n} < \deg B^n.$$

Then the required pair is

$$\boxed{U_{m,n} = R_{m,n}, \quad V_{m,n} = \widehat{V}_{m,n} + A^m Q_{m,n}.}$$

Equivalently, writing $\text{rem}_C(P)$ for the Euclidean remainder of P modulo a nonzero polynomial C ,

$$\boxed{U_{m,n} = \text{rem}_{B^n} \left(U^m \sum_{j=0}^{n-1} \binom{m+j-1}{j} (BV)^j \right),}$$

and then

$$\boxed{V_{m,n} = \frac{1 - A^m U_{m,n}}{B^n}.}$$

The quotient in the last formula is a polynomial, as proved below.

2.1.3 3. The universal binomial Bézout identity

The formula rests on the following identity.

Lemma Let R be any commutative ring, and let $x, y \in R$ satisfy $x + y = 1$. For all $r, s \geq 1$,

$$\boxed{x^r S_{r,s}(y) + y^s S_{s,r}(x) = 1.}$$

Thus explicitly,

$$x^r \sum_{j=0}^{s-1} \binom{r+j-1}{j} y^j + y^s \sum_{i=0}^{r-1} \binom{s+i-1}{i} x^i = 1.$$

Proof For convenience put $S_{r,0} = 0$. Pascal's identity gives, for $r \geq 2$,

$$S_{r,s}(T) - S_{r-1,s}(T) = TS_{r,s-1}(T).$$

Indeed, for $j \geq 1$,

$$\binom{r+j-1}{j} - \binom{r+j-2}{j} = \binom{r+j-2}{j-1}.$$

Since $x = 1 - y$,

$$\begin{aligned} S_{r-1,s}(y) - xS_{r,s}(y) &= S_{r-1,s}(y) - (1-y)S_{r,s}(y) \\ &= S_{r-1,s}(y) - S_{r,s}(y) + yS_{r,s}(y) \\ &= -yS_{r,s-1}(y) + yS_{r,s}(y) \\ &= \binom{r+s-2}{s-1} y^s. \end{aligned}$$

Consequently,

$$x^{r-1}S_{r-1,s}(y) - x^rS_{r,s}(y) = \binom{r+s-2}{s-1} x^{r-1}y^s.$$

On the other hand,

$$S_{s,r}(x) - S_{s,r-1}(x) = \binom{s+r-2}{r-1} x^{r-1}.$$

Because

$$\binom{r+s-2}{s-1} = \binom{r+s-2}{r-1},$$

we obtain

$$x^r S_{r,s}(y) + y^s S_{s,r}(x) = x^{r-1} S_{r-1,s}(y) + y^s S_{s,r-1}(x).$$

Iterating this reduction in r gives

$$x^r S_{r,s}(y) + y^s S_{s,r}(x) = x S_{1,s}(y) + y^s.$$

Finally,

$$S_{1,s}(y) = 1 + y + \cdots + y^{s-1},$$

so

$$x S_{1,s}(y) + y^s = (1 - y)(1 + y + \cdots + y^{s-1}) + y^s 1.$$

This proves the lemma. Since it is an identity with integer coefficients, it remains valid in every characteristic. $\square$

2.1.4 4. Proof of the formula

Apply the lemma with

$$x = AU, \quad y = BV.$$

The given Bézout identity says precisely that $x + y = 1$. Therefore

$$(AU)^m S_{m,n}(BV) + (BV)^n S_{n,m}(AU) = 1.$$

Factoring out A^m and B^n gives

$$A^m (U^m S_{m,n}(BV)) + B^n (V^n S_{n,m}(AU)) = 1,$$

that is,

$$A^m \widehat{U}_{m,n} + B^n \widehat{V}_{m,n} = 1. \tag{1}$$

Thus $(\widehat{U}_{m,n}, \widehat{V}_{m,n})$ is already an explicit Bézout pair for A^m and B^n . It need not, however, satisfy the required degree condition.

Now divide

$$\widehat{U}_{m,n} = B^n Q_{m,n} + R_{m,n}, \quad \deg R_{m,n} < \deg B^n.$$

Substitution into 1 yields

$$\begin{aligned} 1 &= A^m(B^n Q_{m,n} + R_{m,n}) + B^n \widehat{V}_{m,n} \\ &= A^m R_{m,n} + B^n(\widehat{V}_{m,n} + A^m Q_{m,n}). \end{aligned}$$

Hence

$$U_{m,n} = R_{m,n}, \quad V_{m,n} = \widehat{V}_{m,n} + A^m Q_{m,n}$$

has all the required properties.

In particular,

$$B^n \mid 1 - A^m U_{m,n},$$

so

$$V_{m,n} = \frac{1 - A^m U_{m,n}}{B^n}$$

is indeed a polynomial.

2.1.5 5. Uniqueness and classification of all pairs

Suppose that two pairs $((P, Q))$ and $((P', Q'))$ satisfy

$$A^m P + B^n Q = 1, \quad A^m P' + B^n Q' = 1.$$

Subtracting gives

$$A^m(P - P') = -B^n(Q - Q'). \tag{2}$$

Because 1 is itself a Bézout identity for A^m and B^n , equation 2 implies that $B^n \mid P - P'$. Indeed,

$$\begin{aligned} P - P' &= (A^m \widehat{U}_{m,n} + B^n \widehat{V}_{m,n})(P - P') \\ &= \widehat{U}_{m,n} A^m (P - P') + B^n \widehat{V}_{m,n} (P - P'), \end{aligned}$$

and both terms on the right are divisible by B^n .

If B is nonconstant and

$$\deg P, \deg P' < \deg B^n,$$

then either $P - P' = 0$, or

$$\deg(P - P') \geq \deg B^n,$$

which is impossible. Thus $P = P'$, and then 2 gives $Q = Q'$.

If B is a nonzero constant, the degree condition is $\deg P < 0$, hence $P = 0$, so uniqueness is again immediate.

Therefore the normalized pair $(U_{m,n}, V_{m,n})$ is unique.

Moreover, all Bézout pairs without the degree restriction are exactly

$$\boxed{(U_{m,n} + B^n T, V_{m,n} - A^m T), \quad T \in \mathbb{K}[X].}$$

Thus the condition $\deg U_{m,n} < \deg B^n$ selects precisely one pair from this affine family.

2.1.6 6. Symmetric remainder formula

Assume for this subsection that A and B are both nonconstant. Then the normalized pair automatically satisfies

$$\deg V_{m,n} < m \deg A.$$

Indeed, if $V_{m,n} \neq 0$, the highest-degree terms in

$$A^m U_{m,n} + B^n V_{m,n} = 1$$

must cancel, so

$$m \deg A + \deg U_{m,n} = n \deg B + \deg V_{m,n}.$$

Since $\deg U_{m,n} < n \deg B$, it follows that

$$\deg V_{m,n} < m \deg A.$$

Now $V_{m,n} \equiv \widehat{V}_{m,n} \pmod{A^m}$, because

$$V_{m,n} = \widehat{V}_{m,n} + A^m Q_{m,n}.$$

Consequently,

$$\boxed{V_{m,n} = \text{rem}_{A^m} \left(V^n \sum_{i=0}^{m-1} \binom{n+i-1}{i} (AU)^i \right) .}$$

Thus, in the generic nonconstant case, the answer has the completely symmetric form

$$\boxed{\begin{aligned} U_{m,n} &= \text{rem}_{B^n}(U^m S_{m,n}(BV)), \\ V_{m,n} &= \text{rem}_{A^m}(V^n S_{n,m}(AU)). \end{aligned}}$$

2.1.7 7. A directly degree-normalized B -adic formula

The remainder modulo B^n can itself be expanded explicitly using only successive divisions by B .

Assume $\deg B = b \geq 1$. Define

$$\rho_B(P) := \text{rem}_B(P), \quad \delta_B(P) := \frac{P - \rho_B(P)}{B}.$$

Thus

$$P = \rho_B(P) + B\delta_B(P).$$

Iterating gives the finite B -adic expansion

$$P = \sum_{r \geq 0} B^r \rho_B(\delta_B^r(P)).$$

Since

$$\hat{U}_{m,n} = \sum_{k=0}^{n-1} \binom{m+k-1}{k} B^k U^m V^k,$$

its terms of B -adic order strictly below n give

$$\boxed{U_{m,n} = \sum_{r=0}^{n-1} B^r C_r,}$$

where

$$\boxed{C_r = \sum_{k=0}^r \binom{m+k-1}{k}, \rho_B! \left(\delta_B^{r-k}(U^m V^k) \right).}$$

Every C_r has degree $< b$, so the displayed formula directly gives

$$\deg U_{m,n} < nb = \deg B^n.$$

This is equivalent to the single-remainder formula but may be preferable for an implementation based on B -adic digits.

2.1.8 8. Small and boundary cases

Case $m = n = 1$ Here $S_{1,1} = 1$, so

$$\widehat{U}_{1,1} = U, \quad \widehat{V}_{1,1} = V.$$

Because $\deg U < \deg B$, no reduction is needed:

$$(U_{1,1}, V_{1,1}) = (U, V).$$

Case $n = 1$ Since $S_{m,1} = 1$,

$$\boxed{U_{m,1} = \text{rem}_B(U^m)}.$$

The corresponding second coefficient is

$$V_{m,1} = \frac{1 - A^m U_{m,1}}{B}.$$

Before normalization, the symmetric pair is

$$\widehat{U}_{m,1} = U^m, \quad \widehat{V}_{m,1} = V \sum_{i=0}^{m-1} (AU)^i.$$

Case $m = 1$ Because $S_{1,n}(T) = 1 + T + \dots + T^{n-1}$,

$$U_{1,n} = \text{rem}_{B^n} \left(U \sum_{j=0}^{n-1} (BV)^j \right),$$

while

$$\widehat{V}_{1,n} = V^n.$$

The raw identity is simply

$$A, U \sum_{j=0}^{n-1} (BV)^j + (BV)^n = 1.$$

Constant B If $B \in \mathbb{K}^\times$, then $\deg U < 0$ forces $U = 0$, and $AU + BV = 1$ gives

$$V = B^{-1}.$$

The unique answer is therefore

$$\boxed{U_{m,n} = 0, \quad V_{m,n} = B^{-n}.$$

This also covers the possible degenerate case $A = 0$.

Constant A , nonconstant B If $A \in \mathbb{K}^\times$ and B is nonconstant, the canonical initial pair is

$$U = A^{-1}, \quad V = 0.$$

Hence

$$\boxed{U_{m,n} = A^{-m}, \quad V_{m,n} = 0.$$

Positive characteristic There are no exceptional characteristics. The coefficients

$$\binom{m+j-1}{j}$$

are simply reduced modulo $\text{char } \mathbb{K}$. The proof uses no factorial denominators and is an identity over $\mathbb{Z}$.

2.1.9 9. Why the Euclidean reduction is genuinely necessary

The raw pair $(\widehat{U}_{m,n}, \widehat{V}_{m,n})$ need not satisfy the degree condition.

For example, over $\mathbb{Q}$, take

$$A = X^2, \quad B = X + 1, \quad U = 1, \quad V = 1 - X.$$

Then

$$AU + BV = X^2 + (X + 1)(1 - X) = 1.$$

For $m = 1, n = 2$,

$$\widehat{U}_{1,2} = 1 + BV^2 - X^2.$$

But

$$\deg(2 - X^2) = 2 = \deg(X + 1)^2,$$

so the strict degree condition fails. Dividing,

$$2 - X^2 = -(X + 1)^2 + (2X + 3).$$

Thus

$$U_{1,2} = 2X + 3.$$

Also,

$$\widehat{V}_{1,2} = V^2 = (1 - X)^2, \quad Q_{1,2} = -1,$$

so

$$V_{1,2} = (1 - X)^2 - X^2 1 - 2X.$$

Indeed,

$$X^2(2X + 3) + (X + 1)^2(1 - 2X) = 1.$$

2.1.10 10. Hypothesis audit

The identity $AU + BV = 1$ already implies that A and B are coprime, so the explicit coprimality hypothesis is logically redundant once (U, V) are given.

The initial condition $\deg U < \deg B$ is not needed for the construction: the same formula works starting from any Bézout pair $AU + BV = 1$. Its role is to make the given initial pair canonical.

The target condition

$$\deg U_{m,n} < \deg B^n$$

is essential for uniqueness. Without it, all solutions are obtained by adding

$$(B^n T, -A^m T).$$

Finally, the field hypothesis is used for unrestricted Euclidean division in $\mathbb{K}[X]$. The universal binomial identity itself is valid over every commutative ring; the normalized construction also extends, for example, to coefficient

rings over which division by the chosen B is available, such as when B is monic.

Formal verification. Lean source; online type-check.

3 Question Q701

Informal verdict. A — Complete solution, with high confidence.

Lean coverage (partial). Lean proves the main path-pseudometric and finite-chain characterization of extreme points, convexity, and two illustrative examples. The optional McShane-extension, exposed-point, strong-extremality, and alpha-equals-one discussions are not formalized.

3.1 Audited informal answer

3.1.1 1. Faithful restatement

Fix an exponent $\alpha \in (0, 1)$. Let

$$\mathcal{C}_\alpha = \{f : [0, 1] \rightarrow \mathbb{R} : f(0) = 0, \quad |f(x) - f(y)| \leq |x - y|^\alpha \text{ for all } x, y \in [0, 1]\}.$$

Equipped with the uniform topology, $\mathcal{C}_\alpha$ is a compact convex subset of $C([0, 1])$. The problem asks for simple properties and, ideally, a characterization of its extreme points. There are no subparts or additional qualifications in the PDF.

The phrase “ α -Hölder of ratio 1” must mean Hölder constant **at most** 1. If it meant that the optimal constant were exactly 1, the set would not be convex: x^α and $-x^\alpha$ would belong to it, whereas their midpoint 0 would not.

The answer below gives a complete intrinsic characterization.

3.1.2 2. The slack pseudometric

Write

$$d_\alpha(x, y) = |x - y|^\alpha.$$

Since $0 < \alpha < 1$, the inequality

$$(r + s)^\alpha \leq r^\alpha + s^\alpha$$

shows that d_α is a metric.

For $f \in \mathcal{C}_\alpha$, define its **slack** on a pair $((x,y))$ by

$$\sigma_f(x, y) == d_\alpha(x, y) - |f(x) - f(y)||x - y|^\alpha - |f(x) - f(y)|.$$

Thus $\sigma_f(x, y) \geq 0$, and $\sigma_f(x, y) = 0$ precisely when the Hölder inequality is saturated at $((x,y))$.

For a finite chain

$$P = (x_0, x_1, \dots, x_n)$$

define its total slack by

$$\ell_f(P) = \sum_{i=1}^n \sigma_f(x_{i-1}, x_i).$$

Finally, define

$$\rho_f(x, y) = \inf \{ \ell_f(x_0, \dots, x_n) : n \geq 1, x_0 = x, x_n = y \},$$

and put $\rho_f(x, x) = 0$.

The function ρ_f is a pseudometric. Indeed, symmetry follows by reversing chains, and the triangle inequality follows by concatenating chains. Moreover, using the one-edge chain,

$$0 \leq \rho_f(x, y) \leq \sigma_f(x, y) \leq d_\alpha(x, y). \quad (2.1)$$

It is also the largest pseudometric bounded above by σ_f .

3.1.3 3. Complete characterization

Theorem For $f \in \mathcal{C}_\alpha$, the following conditions are equivalent.

1. f is an extreme point of $\mathcal{C}_\alpha$.
1. For every $x \in [0, 1]$,

$$\rho_f(0, x) = 0. \quad (3.1)$$

1. For every $x, y \in [0, 1]$,

$$\rho_f(x, y) = 0. \quad (3.2)$$

1. For every $x \in [0, 1]$ and every $\varepsilon > 0$, there exist an integer $n \geq 1$ and points

$$0 = x_0, x_1, \dots, x_n = x$$

such that

$$\sum_{i=1}^n (|x_i - x_{i-1}|^\alpha - |f(x_i) - f(x_{i-1})|) < \varepsilon.$$

(3.3)

Equivalently, f is extreme precisely when every point can be joined to 0 by finite chains along which the accumulated deficit in the Hölder inequalities is arbitrarily small.

A general pointed-metric-space version of this criterion appears in a theorem of Jeff D. Farmer; the proof below is self-contained. ([SciSpace][1])

3.1.4 4. Proof of the characterization

We first identify all possible midpoint perturbations of f .

Lemma 4.1: midpoint perturbations Let $u : [0, 1] \rightarrow \mathbb{R}$ satisfy $u(0) = 0$. Then

$$f + u \in \mathcal{C}_\alpha \quad \text{and} \quad f - u \in \mathcal{C}_\alpha$$

if and only if

$$|u(x) - u(y)| \leq \sigma_f(x, y) \quad \text{for all } x, y \in [0, 1]. \quad (4.1)$$

Proof For real numbers (A, B) ,

$$\max |A + B|, |A - B| = |A| + |B|. \quad (4.2)$$

Indeed, if A and B have the same sign, the first absolute value equals $(|A| + |B|)$; if they have opposite signs, the second one does.

Apply this with

$$A = f(x) - f(y), \quad B = u(x) - u(y).$$

The two conditions

$$|A + B| \leq d_\alpha(x, y), \quad |A - B| \leq d_\alpha(x, y)$$

are therefore equivalent to

$$|A| + |B| \leq d_\alpha(x, y),$$

which is exactly

$$|u(x) - u(y)| \leq d_\alpha(x, y) - |f(x) - f(y)| = \sigma_f(x, y).$$

This proves the lemma. $\square$

Lemma 4.2: path formulation Condition (4.1) is equivalent to

$$|u(x) - u(y)| \leq \rho_f(x, y) \quad \text{for all } x, y. \quad (4.3)$$

Proof If (4.1) holds, then for every chain $x = x_0, \dots, x_n = y$,

$$\begin{aligned} |u(x) - u(y)| &\leq \sum_{i=1}^n |u(x_i) - u(x_{i-1})| \\ &\leq \sum_{i=1}^n \sigma_f(x_{i-1}, x_i). \end{aligned}$$

Taking the infimum over all chains gives (4.3).

Conversely, (2.1) gives $\rho_f(x, y) \leq \sigma_f(x, y)$, so (4.3) implies (4.1). $\square$

Thus all symmetric decompositions

$$f = \frac{(f + u) + (f - u)}{2}$$

are described exactly by the 1-Lipschitz functions for the pseudometric ρ_f , anchored at 0.

Proof that 2 implies extremality Suppose that $\rho_f(0, x) = 0$ for every x .

Assume

$$f = \frac{g+h}{2}, \quad g, h \in \mathcal{C}_\alpha.$$

Set

$$u = \frac{g-h}{2}.$$

Then $g = f + u$, $h = f - u$, and $u(0) = 0$. By Lemma 4.2,

$$|u(x)| = |u(x) - u(0)| \leq \rho_f(0, x) = 0.$$

Hence $u = 0$, so $g = h = f$. Therefore f is extreme.

Proof that failure of 2 implies non-extremality Suppose that

$$\rho_f(0, x_*) > 0$$

for some $x_* \in [0, 1]$. Define

$$u(x) = \rho_f(0, x).$$

The reverse triangle inequality for the pseudometric ρ_f gives

$$|u(x) - u(y)| = |\rho_f(0, x) - \rho_f(0, y)| \leq \rho_f(x, y).$$

By (2.1),

$$|u(x) - u(y)| \leq \rho_f(x, y) \leq \sigma_f(x, y).$$

Consequently, Lemma 4.1 gives

$$f + u, f - u \in \mathcal{C}_\alpha.$$

Moreover, $u(0) = 0$, while $u(x_*) > 0$, so $u \neq 0$. Thus

$$f = \frac{(f+u) + (f-u)}{2}$$

is a nontrivial midpoint decomposition, and f is not extreme.

This proves the equivalence of 1 and 2.

Conditions 2 and 3 are equivalent because

$$\rho_f(x, y) \leq \rho_f(x, 0) + \rho_f(0, y).$$

Finally, 2 and 4 are equivalent by the definition of the infimum $\rho_f(0, x)$. The theorem is proved. $\square$

3.1.5 5. Quotient-space form of the answer

Define an equivalence relation by

$$x \sim_f y \iff \rho_f(x, y) = 0.$$

The pseudometric ρ_f induces a genuine metric on the quotient

$$X_f = [0, 1] / \sim_f.$$

Every nontrivial decomposition of f is obtained by choosing a nonzero anchored 1-Lipschitz function v on X_f , pulling it back to $u = v \circ q_f$, and writing

$$f = \frac{(f + u) + (f - u)}{2}.$$

Hence another exact formulation is

$f \text{ is extreme} \iff X_f \text{ consists of a single point.}$

This not only determines whether f is extreme; it describes all its possible symmetric midpoint decompositions.

3.1.6 6. Saturated pairs and a simple sufficient condition

Call $(\{x, y\})$ a **tight pair** if

$$|f(x) - f(y)| = |x - y|^\alpha.$$

Let G_f be the graph with vertex set $([0, 1])$ and an edge between every tight pair.

If every point can be connected to 0 by a finite path in G_f , then every edge in the path has zero slack, so $\rho_f(0, x) = 0$. Thus f is extreme.

More generally, let R_f be the set of points connected to 0 by finite tight paths. If R_f is dense in $([0, 1])$, then f is extreme. Indeed, $x \mapsto \rho_f(0, x)$ is α -Hölder by

$$|\rho_f(0, x) - \rho_f(0, y)| \leq \rho_f(x, y) \leq |x - y|^\alpha,$$

and it vanishes on the dense set R_f .

A stronger conclusion: exposed points If R_f is dense, then f is in fact an **exposed point** of $\mathcal{C}_\alpha$.

To see this, choose a dense sequence $(z_n) \subset R_f$. For each n , choose a finite tight path from 0 to z_n . Enumerate all its oriented edges as (p_m, q_m) , orienting each one so that

$$\varepsilon_m(f(q_m) - f(p_m)) = |q_m - p_m|^\alpha, \quad \varepsilon_m \in -1, 1.$$

Define the continuous linear functional

$$\Lambda(g) = \sum_{m=1}^{\infty} 2^{-m} \varepsilon_m(g(q_m) - g(p_m)).$$

For every $g \in \mathcal{C}_\alpha$,

$$\Lambda(g) \leq \sum_{m=1}^{\infty} 2^{-m} |q_m - p_m|^\alpha = \Lambda(f).$$

Equality forces equality on every oriented tight edge, hence $g(z_n) = f(z_n)$ for all n . Continuity and density then give $g = f$. Thus f is the unique maximizer of Λ .

3.1.7 7. The finite-set classification

Let

$$F = t_0, t_1, \dots, t_N, \quad t_0 = 0,$$

be finite, and let $\mathcal{C}_\alpha(F)$ be the analogous finite-dimensional polytope.

For $f \in \mathcal{C}_\alpha(F)$, form the graph whose edges are precisely the saturated pairs

$$|f(t_i) - f(t_j)| = |t_i - t_j|^\alpha.$$

Then

$f \in \text{Ext } \mathcal{C}_\alpha(F) \iff \text{the saturated-pair graph is connected.}$
--

Indeed, in a finite graph the zero weighted-path distance from 0 to every vertex is equivalent to the existence of a path made entirely of zero-weight edges. If no such path exists, the minimum over the finitely many simple paths is strictly positive.

Equivalently, a finite extreme point is characterized by the existence of a spanning tree all of whose edge constraints are saturated.

This yields an exact $O(N^2)$ recognition algorithm: construct all saturated edges and run a graph search from t_0 .

3.1.8 8. Explicit extreme points

8.1 Distance cones For $p \in [0, 1]$, define

$$k_p(x) = p^\alpha - |x - p|^\alpha.$$

Then $k_p(0) = 0$. Since d_α is a metric,

$$|k_p(x) - k_p(y)| = |d_\alpha(p, x) - d_\alpha(p, y)| \leq d_\alpha(x, y),$$

so $k_p \in \mathcal{C}_\alpha$.

Moreover,

$$k_p(p) - k_p(x) = |x - p|^\alpha.$$

Thus every x is tightly connected to p , and p is tightly connected to 0. Therefore k_p is extreme, indeed exposed. The same holds for $-k_p$.

Special cases include

$$k_0(x) = -x^\alpha, \quad -k_0(x) = x^\alpha,$$

and

$$k_1(x) = 1 - (1 - x)^\alpha.$$

8.2 A one-turn family For $c \in [0, 1]$, define

$$T_c(x) = \begin{cases} x^\alpha, & 0 \leq x \leq c, \\ c^\alpha - (x - c)^\alpha, & c \leq x \leq 1. \end{cases} \quad (8.1)$$

We verify that $T_c \in \mathcal{C}_\alpha$.

If both points lie on the same side of c , this follows from

$$|r^\alpha - s^\alpha| \leq |r - s|^\alpha.$$

Now let $x \leq c \leq y$. Then

$$\begin{aligned} T_c(x) - T_c(y) &= x^\alpha - c^\alpha + (y - c)^\alpha \\ &\leq (y - c)^\alpha \leq (y - x)^\alpha, \end{aligned}$$

while

$$\begin{aligned} T_c(y) - T_c(x) &= c^\alpha - x^\alpha - (y - c)^\alpha \\ &\leq c^\alpha - x^\alpha \leq (c - x)^\alpha \leq (y - x)^\alpha. \end{aligned}$$

Hence $|T_c(x) - T_c(y)| \leq |x - y|^\alpha$.

Every $x \leq c$ is tightly connected to 0, since $T_c(x) = x^\alpha$. Every $x \geq c$ is tightly connected to c , since

$$T_c(c) - T_c(x) = (x - c)^\alpha.$$

Thus T_c is extreme and exposed.

3.1.9 9. McShane-type construction of further extreme points

Let $A \subset [0, 1]$ be compact, with $0 \in A$, and let $g : A \rightarrow \mathbb{R}$ be an extreme anchored 1-Lipschitz function for d_α .

Define the two canonical extensions

$$M^+g(x) = \inf_{y \in A} (g(y) + |x - y|^\alpha),$$

and

$$M^-g(x) = \sup_{y \in A} (g(y) - |x - y|^\alpha).$$

Both are elements of $\mathcal{C}_\alpha$ and agree with g on A .

Because A is compact, the infimum or supremum is attained. Thus for every x , there is some $y_x \in A$ such that either

$$M^+g(x) - g(y_x) = |x - y_x|^\alpha$$

or

$$g(y_x) - M^-g(x) = |x - y_x|^\alpha.$$

Consequently x is joined to A by a zero-slack edge. Within A , extremality of g supplies arbitrarily small-slack chains to 0. Concatenating them proves, by the main theorem, that both M^+g and M^-g are extreme.

In particular, beginning with a finite extreme configuration—equivalently, a connected saturated graph—and applying either extension gives a large explicit class of extreme functions on the whole interval.

3.1.10 10. Necessary conditions, and why they are not sufficient

Let

$$[f]_\alpha = \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\alpha}.$$

Every extreme point lies on the Hölder sphere If $[f]_\alpha = L < 1$, set

$$u(x) = (1 - L)x^\alpha.$$

Then $[u]_\alpha = 1 - L$, and therefore

$$[f \pm u]_\alpha \leq [f]_\alpha + [u]_\alpha = 1.$$

Thus $f \pm u \in \mathcal{C}_\alpha$, $u \neq 0$, and f is not extreme. Hence

$$f \in \text{Ext } \mathcal{C}_\alpha \implies [f]_\alpha = 1. \tag{10.1}$$

Hölder norm 1 is not sufficient Consider

$$f(x) = x.$$

Because $0 \leq |x - y| \leq 1$,

$$|x - y| \leq |x - y|^\alpha,$$

so $f \in \mathcal{C}_\alpha$, and $[f]_\alpha = 1$, with equality at $((0,1))$.

Nevertheless f is not extreme. Define

$$u(x) = (1 - \alpha)x(1 - x).$$

For $x < y$, put $r = y - x$. Then

$$|u(y) - u(x)| = (1 - \alpha)r, |1 - x - y|.$$

Since $0 \leq x \leq 1 - r$,

$$|1 - x - y| \leq 1 - r,$$

and hence

$$|u(y) - u(x)| \leq (1 - \alpha)r(1 - r).$$

The convexity inequality

$$r^{-(1-\alpha)} \geq 1 + (1 - \alpha)(1 - r)$$

gives, after multiplication by r ,

$$r^\alpha - r \geq (1 - \alpha)r(1 - r).$$

Therefore

$$|u(y) - u(x)| \leq r^\alpha - r = |x - y|^\alpha - |f(x) - f(y)|.$$

Lemma 4.1 now yields

$$f + u, \quad f - u \in \mathcal{C}_\alpha.$$

Since $u \neq 0$, f is not extreme.

Thus the condition $[f]_\alpha = 1$ is necessary but far from sufficient.

3.1.11 11. Strong extremality

A point $f \in \mathcal{C}_\alpha$ is called strongly extreme in the uniform norm if

$$\left\| \frac{g_n + h_n}{2} - f \right\|_\infty \longrightarrow 0 \implies |g_n - h_n| * \infty \longrightarrow 0$$

for all sequences $g_n, h_n \in \mathcal{C}_\alpha$.

Because $\mathcal{C}_\alpha$ is compact in the uniform norm,

$$\boxed{f \text{ is extreme} \iff f \text{ is strongly extreme.}}$$

Only the nontrivial implication needs proof. If it failed, there would be sequences with midpoint converging to f but

$$|g_n - h_n|_\infty \geq \varepsilon > 0.$$

By compactness, a subsequence would satisfy

$$g_n \rightarrow g, \quad h_n \rightarrow h$$

uniformly. Then

$$f = \frac{g+h}{2}, \quad |g-h|_\infty \geq \varepsilon,$$

contradicting extremality.

3.1.12 12. Boundary cases and necessity of the hypotheses

The case $\alpha = 1$ The same slack-pseudometric theorem remains valid. It reduces to the classical concrete characterization

$$f \in \text{Ext } \mathcal{C}_1 \iff |f'(t)| = 1 \text{ for almost every } t \in [0, 1].$$

Here every 1-Lipschitz function is absolutely continuous and $|f'| \leq 1$ almost everywhere.

Indeed, for $x < y$,

$$\rho_f(x, y) = \int_x^y (1 - |f'(t)|) dt. \quad (12.1)$$

For the upper bound, take increasingly fine monotone partitions. The sum of the absolute increments converges upward to the total variation

$$\text{Var}_{[x, y]}(f) = \int_x^y |f'(t)| dt.$$

For the reverse inequality, every edge $[z_{i-1}, z_i]$ of an arbitrary chain satisfies

$$|z_i - z_{i-1}| - |f(z_i) - f(z_{i-1})| \geq \int_{[z_{i-1}, z_i]} (1 - |f'|).$$

Every point between x and y is crossed by at least one edge of a chain from x to y , so summation gives the reverse inequality in (12.1).

Notice the sharp change at the boundary: $f(x) = x$ is extreme for $\alpha = 1$, but, as shown above, is not extreme for any $0 < \alpha < 1$.

The case $\alpha > 1$ Every such Hölder function on an interval is constant. Indeed, subdividing $([x,y])$ into n equal pieces gives

$$|f(y) - f(x)| \leq n \left(\frac{y-x}{n} \right)^\alpha = (y-x)^\alpha n^{1-\alpha} \rightarrow 0.$$

With $f(0) = 0$, the set consists only of the zero function, which is trivially extreme.

The case $\alpha = 0$ The stated compactness fails. Even among continuous functions, the family with oscillation at most 1 is not equicontinuous; for example,

$$f_n(x) = \min(1, nx)$$

has no uniformly convergent subsequence.

The anchoring condition The condition $f(0) = 0$ is essential. Without it, arbitrary constants may be added:

$$f = \frac{(f+c) + (f-c)}{2}.$$

Thus the unanchored set has no extreme points and is not bounded in the uniform norm.

The real-valued condition The scalar identity

$$\max(|A+B|, |A-B|) = |A| + |B|$$

is specifically real. For complex- or vector-valued Hölder maps, the midpoint perturbation condition and therefore the characterization require modification.

3.1.13 13. Compactness, separation, countability, and choice audit

The ambient space is the real Banach space $C([0,1])$ with the uniform norm, hence a Hausdorff metric space.

For every $f \in \mathcal{C}_\alpha$,

$$|f(x)| \leq x^\alpha \leq 1,$$

and the family has the common modulus of continuity $r \mapsto r^\alpha$. It is closed under uniform convergence. The metric form of Arzelà–Ascoli therefore gives compactness. In this setting compactness and sequential compactness are equivalent.

The main extremality theorem itself is algebraic and metric; it uses neither compactness nor separability and works for arbitrary pointed metric spaces. Compactness is used only for the strong-extremality consequence and for attainment in the extension construction.

No general net argument, transfinite induction, or nonconstructive choice principle is required. The exposed-point argument uses only the separability of $([0,1])$, through a countable dense sequence.

3.1.14 Final classification

The extreme points of the compact convex set in Q701 are exactly the functions satisfying

$$\forall x \in [0, 1] \ \forall \varepsilon > 0 \exists, 0 = x_0, x_1, \dots, x_n = x \quad \sum_{i=1}^n (|x_i - x_{i-1}|)^\alpha - \varepsilon$$

Equivalently, the largest pseudometric lying below the pairwise Hölder slack

$$|x - y|^\alpha - |f(x) - f(y)|$$

must vanish identically. This criterion is necessary and sufficient and also gives a canonical nonzero perturbation $u(x) = \rho_f(0, x)$ whenever f is not extreme.

[1]: <https://scispace.com/pdf/extreme-points-of-the-unit-ball-of-the-space-of-lipschitz-3l772juamw.pdf> "Extreme points of the unit ball of the space of Lipschitz functions"

Formal verification. Lean source; online type-check.

4 Question Q668

Informal verdict. C — Substantial progress, with high confidence in the central classification.

Lean coverage (partial). Lean proves the structural periodic-point and asymptotic classification, the power-of-two criterion, the complete binary 15-cycle for $n=5$, and the explicit eigenray tending to zero. The Section 6 example and the general basin-of-attraction problem are not formalized or claimed solved.

4.1 Audited informal answer

4.2 Partial result

I give below:

1. a complete classification of all possible periodic attractors for every n ;
2. a proof that every real orbit is asymptotic, phase by phase, to one of them;
3. an exact finite algorithm for rational data;
4. a complete description of the possible asymptotic cycles for $n = 5$;
5. an explicit nonconstant 5-tuple converging to 0;
6. an exact algebraic 5-tuple which approaches the nonzero 15-cycle but never enters any periodic orbit.

What remains unresolved by this analysis is a direct criterion, for an arbitrary exact real X , deciding which attractor basin contains X and computing the limiting scale without iterating indefinitely.

4.2.1 1. Faithful restatement

Let $n \geq 3$. For

$$X = (x_1, \dots, x_n) \in \mathbb{R}_+^n,$$

define

$$P(X) = (|x_1 - x_2|, |x_2 - x_3|, \dots, |x_{n-1} - x_n|, |x_n - x_1|).$$

Study the orbit

$$X, P(X), P^2(X), \dots$$

for a prescribed starting vector X .

The editorial note says that $n = 4$ is understood: every orbit eventually reaches 0, except for one algebraic family, up to elementary transformations, represented after one iteration by

$$(1, a, a^2, a^3),$$

where $a - 1$ is the real root of

$$u^3 + 2u^2 - 2 = 0.$$

Equivalently, a satisfies

$$a^3 - a^2 - a - 1 = 0.$$

The note explicitly welcomes any partial result encompassing $n = 5$. There are two minor notational points.

- Here $\mathbb{R}_+$ must mean $[0, \infty)$, since $P(X)$ can have zero coordinates.
- The phrase “up to a simple transformation” in the editorial note is informal and does not specify the transformations, but this does not affect the formal problem.

Throughout, write

$$X^{(q)} = P^q(X), \quad M_q = |X^{(q)}|_\infty.$$

4.3 2. Main classification theorem

Write

$$n = hm, \quad h = 2^{\nu_2(n)}, \quad m \text{ odd}.$$

Thus h is the largest power of 2 dividing n .

For a binary vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in 0, 1^n$, put

$$E_\varepsilon(z) = \sum_{j=1}^n \varepsilon_j z^{j-1} \in \mathbb{F}_2[z].$$

Define

$$\mathcal{S}_n = \left\{ \varepsilon \in 0, 1^n : 1 + z^h \mid E_\varepsilon(z) \text{ in } \mathbb{F}_2[z] \right\}.$$

Equivalently,

$$\varepsilon \in \mathcal{S}_n \iff \sum_{k=0}^{m-1} \varepsilon_{r+kh} \equiv 0 \pmod{2} \quad (1 \leq r \leq h). \quad (2.1)$$

Thus $\mathcal{S}_n$ is given by h independent parity conditions and

$$|\mathcal{S}_n| = 2^{n-h}.$$

4.3.1 Theorem 2.1

For every $X \in \mathbb{R}_+^n$:

1. The sequence M_q is nonincreasing, and therefore

$$c(X) := \lim_{q \rightarrow \infty} M_q$$

exists.

1. There exists a periodic binary sequence $(\varepsilon^{(q)})_{q \geq 0}$, with every $\varepsilon^{(q)} \in \mathcal{S}_n$ and

$$\varepsilon^{(q+1)} = P(\varepsilon^{(q)}),$$

such that

$$\lim_{q \rightarrow \infty} \left| P^q(X) - c(X) \varepsilon^{(q)} \right|_{\infty} = 0. \quad (2.2)$$

1. The periodic points of P in $\mathbb{R}_+^n$ are exactly

$$c\varepsilon, \quad c \geq 0, \quad \varepsilon \in \mathcal{S}_n. \quad (2.3)$$

1. The map P permutes the finite set $\mathcal{S}_n$. Hence all possible periodic cycles can be computed by taking the cycle decomposition of this permutation.
1. Every real orbit converges to 0 if and only if n is a power of 2.

An equivalent limit-point and periodic-orbit classification appears in the classical work of Misiurewicz and Schinzel; the proof below is self-contained and also makes the phase-by-phase convergence explicit. ([hrj.epi-sciences.org][1])

4.4 3. Proof of Theorem 2.1

4.4.1 3.1. Monotonicity

If $Y = (y_1, \dots, y_n) \in [0, M]^n$, then

$$|y_i - y_{i+1}| \leq M.$$

Consequently,

$$|P(Y)| * \infty \leq |Y| * \infty.$$

Thus $M_{q+1} \leq M_q$, proving the existence of $c(X)$.

We shall need the following strict form.

4.4.2 3.2. Strict decrease unless all entries are extremal

For $u_0, \dots, u_k \in [0, M]$, define recursively

$$\Delta_0(u_0) = u_0$$

and

$$\Delta_k(u_0, \dots, u_k) = |\Delta_{k-1}(u_0, \dots, u_{k-1}) - \text{-----}|$$

Then, with cyclic indices,

$$(P^k(Y))_i = \Delta_k(y_i, y_{i+1}, \dots, y_{i+k}). \quad (3.1)$$

Lemma 3.1 Suppose $u_0, \dots, u_k \in [0, M]$. If

$$\Delta_k(u_0, \dots, u_k) = M,$$

then

$$u_j \in 0, M \quad (0 \leq j \leq k).$$

Proof First observe the following endpoint fact.

If

$$u_0, \dots, u_{k-1} \in 0, M, \quad u_k = t \in [0, M],$$

then

$$\Delta_k(u_0, \dots, u_k) \in t, M - t. \quad (3.2)$$

Indeed, after taking one row of absolute differences, all entries except possibly the last are again in $(\{0, M\})$, while the last is either t or $M - t$. Induction proves (3.2). The analogous assertion holds when the exceptional entry is the first one.

We now prove the lemma by induction on k . The case $k = 0$ is immediate. Set

$$A = \Delta_{k-1}(u_0, \dots, u_{k-1}), \quad B = \Delta_{k-1}(u_1, \dots, u_k).$$

Both A and B belong to $([0, M])$. Since

$$|A - B| = M,$$

we must have

$$A, B = 0, M.$$

Suppose, for example, that $A = M$. By the induction hypothesis,

$$u_0, \dots, u_{k-1} \in 0, M.$$

Since $B = 0$, the endpoint observation applied to $u_1, \dots, u_k$ implies that $u_k \in 0, M$. The other case is symmetric. $\square$

Corollary 3.2 If $Y \in [0, M]^n$ has at least one coordinate strictly between 0 and M , then

$$\boxed{|P^{n-1}(Y)|_\infty < M.} \quad (3.3)$$

Proof Every coordinate of $P^{n-1}(Y)$ is an $(n - 1)$ -st absolute difference involving all n original coordinates once. Lemma 3.1 shows that none of these coordinates can equal M . $\square$

4.4.3 3.3. Every limit point is binary up to scale

Let $\Omega(X)$ be the set of limit points of the orbit.

Take $Y \in \Omega(X)$. There is a sequence $q_j \rightarrow \infty$ such that

$$X^{(q_j)} \longrightarrow Y.$$

Since $M_{q_j} \rightarrow c(X)$,

$$|Y|_\infty = c(X).$$

Suppose $c(X) > 0$ and that some coordinate of Y lies strictly between 0 and $c(X)$. Corollary 3.2 gives

$$|P^{n-1}(Y)|_\infty < c(X).$$

On the other hand, continuity of P yields

$$P^{n-1}(X^{(q_j)}) \longrightarrow P^{n-1}(Y),$$

while

$$\left\| P^{n-1}(X^{(q_j)}) \right\|_\infty = M_{q_j+n-1} \longrightarrow c(X),$$

a contradiction.

Therefore every limit point has the form

$$Y = c(X)\varepsilon, \quad \varepsilon \in 0, 1^n. \tag{3.4}$$

When $c(X) = 0$, the orbit itself converges to 0.

4.4.4 3.4. Algebra over $\mathbb{F}_2$

On binary vectors, absolute difference agrees with addition modulo 2:

$$|a - b| = a + b \pmod{2} \quad (a, b \in 0, 1).$$

Let $L : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ denote the resulting linear map

$$(L\varepsilon)_i = \varepsilon_i + \varepsilon_{i+1}.$$

Identify $\mathbb{F}_2^n$ with

$$R_n = \mathbb{F}_2[z]/(z^n - 1)$$

by $\varepsilon \mapsto E_\varepsilon(z)$. Then

$$E_{L\varepsilon}(z) = z^{-1}(1+z)E_\varepsilon(z) \quad \text{in } R_n. \quad (3.5)$$

Since multiplication by z is invertible in R_n , the image of L^h is the ideal generated by

$$(1+z)^h = 1+z^h,$$

where we used that h is a power of 2.

Moreover $1+z^h \mid z^n - 1$ in $\mathbb{F}_2[z]$. It follows that

$$\text{im } L^h = \mathcal{S}_n. \quad (3.6)$$

Reducing E_ε modulo $1+z^h$ gives

$$E_\varepsilon(z) \equiv \sum_{r=1}^h \left(\sum_{k=0}^{m-1} \varepsilon_{r+kh} \right) z^{r-1},$$

which proves the parity formulation (2.1).

4.4.5 3.5. Why limit points belong to $\mathcal{S}_n$

The omega-limit set satisfies

$$P^k(\Omega(X)) = \Omega(X) \quad (k \geq 1). \quad (3.7)$$

Indeed, the inclusion from left to right follows by continuity. Conversely, if

$$Y = \lim_j X^{(q_j)},$$

take a convergent subsequence of $X^{(q_j-k)}$; its limit $Z \in \Omega(X)$ satisfies $P^k(Z) = Y$.

Now let $Y = c\varepsilon \in \Omega(X)$, with $c > 0$. By (3.7), there exists $c\delta \in \Omega(X)$ such that

$$P^h(c\delta) = c\varepsilon.$$

Hence

$$\varepsilon = L^h \delta \in \text{im } L^h = \mathcal{S}_n.$$

Thus

$$\Omega(X) \subseteq c(X)\mathcal{S}_n. \quad (3.8)$$

4.4.6 3.6. P is a permutation of $\mathcal{S}_n$

Since

$$\mathcal{S}_n = \text{im } L^h,$$

we have $L(\mathcal{S}_n) \subseteq \mathcal{S}_n$.

It remains to prove injectivity. The kernel of L consists of the two constant binary vectors:

$$\ker L = 0, \mathbf{1}, \quad \mathbf{1} = (1, \dots, 1).$$

We claim that $\mathbf{1} \notin \mathcal{S}_n$. Its polynomial is

$$J_n(z) = 1 + z + \dots + z^{n-1} = \frac{z^n + 1}{z + 1} \quad \text{in characteristic 2.}$$

Because $n = hm$ with m odd,

$$z^n + 1 = (z^m + 1)^h.$$

The root $z = 1$ has multiplicity exactly one in $z^m + 1$, since

$$\left. \frac{d}{dz}(z^m + 1) \right|_{z=1} = m \equiv 1 \pmod{2}.$$

Thus $z = 1$ has multiplicity h in $z^n + 1$, and multiplicity $h - 1$ in J_n . Therefore

$$(1 + z)^h = 1 + z^h \nmid J_n(z).$$

So $\mathbf{1} \notin \mathcal{S}_n$.

If $L\varepsilon = L\delta$ with $\varepsilon, \delta \in \mathcal{S}_n$, then

$$\varepsilon + \delta \in \ker L.$$

It cannot equal $\mathbf{1}$, because $\mathcal{S}_n$ is a vector space and $\mathbf{1} \notin \mathcal{S}_n$. Hence $\varepsilon = \delta$.

Thus L , and therefore P , permutes $\mathcal{S}_n$.

It also follows that the binary periodic points are precisely the elements of $\mathcal{S}_n$. Indeed:

- every element of $\mathcal{S}_n$ is periodic because L permutes this finite set;
- if $L^r \varepsilon = \varepsilon$, choose a multiple $s \geq h$ of r ; then

$$\varepsilon = L^s \varepsilon = L^h (L^{s-h} \varepsilon),$$

so $\varepsilon \in \text{im } L^h = \mathcal{S}_n$.

4.4.7 3.7. The orbit shadows one periodic cycle

We know that all limit points lie in the finite set

$$F = c(X)\mathcal{S}_n.$$

Consequently,

$$\text{dist}(X^{(q)}, F) \longrightarrow 0. \quad (3.9)$$

Otherwise a subsequence staying a positive distance from F would have a limit point outside F .

Assume $c(X) > 0$. Distinct points of F are separated in the supremum norm by at least $c(X)$.

The map P is 2-Lipschitz:

$$|P(U) - P(V)| * \infty \leq 2|U - V| * \infty. \quad (3.10)$$

For all sufficiently large q , there is therefore a unique $f_q \in F$ with

$$|X^{(q)} - f_q|_\infty < \frac{c(X)}{10}.$$

Furthermore,

$$\begin{aligned} |P(f_q) - f_{q+1}| * \infty &\leq |P(f_q) - P(X^{(q)})| * \infty + |X^{(q+1)} - f_{q+1}|_\infty \\ &< \frac{3c(X)}{10}. \end{aligned}$$

Since two distinct elements of F are at distance at least $c(X)$, this forces

$$f_{q+1} = P(f_q).$$

Hence, from some point onward, the nearest elements f_q follow one periodic cycle. Together with (3.9), this proves (2.2).

4.4.8 3.8. All real periodic points

Let Y be periodic. The sequence $|P^q(Y)|_\infty$ is both periodic and nonincreasing, hence constant.

If some coordinate of Y were strictly between 0 and $M = |Y|_\infty$, Corollary 3.2 would give

$$|P^{n-1}(Y)|_\infty < M,$$

a contradiction. Thus

$$Y = M\varepsilon$$

for a binary vector ε . The binary classification proves $\varepsilon \in \mathcal{S}_n$.

Conversely, $M\varepsilon$ is periodic whenever $\varepsilon \in \mathcal{S}_n$, by positive homogeneity:

$$P(M\varepsilon) = MP(\varepsilon).$$

This proves (2.3).

Finally, if n is a power of 2, then $h = n$, so the only degree- $< n$ polynomial divisible by $1 + z^n$ is zero. Hence

$$\mathcal{S}_n = 0$$

and every orbit converges to 0.

If n is not a power of 2, then $h < n$, so

$$|\mathcal{S}_n| = 2^{n-h} > 1,$$

and nonzero periodic orbits exist. $\square$

4.5 4. Complete asymptotic description for $n = 5$

For $n = 5$,

$$h = 1, \quad m = 5.$$

Condition (2.1) becomes

$$\varepsilon_1 + \cdots + \varepsilon_5 \equiv 0 \pmod{2}.$$

Thus

$$\boxed{\mathcal{S}_5 = \{\varepsilon \in \{0, 1\}^5 : \varepsilon \text{ has even Hamming weight}\}} \quad (4.1)$$

There are 16 such vectors.

4.5.1 Proposition 4.1

The zero vector is fixed, and the other 15 vectors of $\mathcal{S}_5$ form one cycle of exact length 15.

Proof Set

$$H(z) = 1 + z + z^2 + z^3 + z^4.$$

This polynomial is irreducible over $\mathbb{F}_2$:

- it has no root in $\mathbb{F}_2$;
- if it factored without a linear factor, it would be a product of two irreducible quadratics;
- the only monic irreducible quadratic over $\mathbb{F}_2$ is

$z^2 + z + 1$, whose square is $z^4 + z^2 + 1 \neq H(z)$.

Consequently,

$$K = \mathbb{F}_2[z]/(H)$$

is the field with 16 elements.

The space $\mathcal{S}_5 = (1 + z)R_5$ is naturally isomorphic, as a vector space, to K . Under this identification, the binary Ducci map acts as multiplication by

$$\mu = 1 + \alpha,$$

where α is a nontrivial fifth root of unity in K .

Since

$$1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0,$$

we get

$$\mu^3 = (1 + \alpha)^3 = 1 + \alpha + \alpha^2 + \alpha^3 = \alpha^4.$$

Thus μ^3 has order 5. Since $K^\times$ has order 15, the order of μ is either 5 or 15.

But

$$\mu^5 = (1 + \alpha)^5 = (1 + \alpha^4)(1 + \alpha) = \alpha + \alpha^4.$$

If $\mu^5 = 1$, then

$$1 + \alpha + \alpha^4 = 0.$$

Adding $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0$ gives

$$\alpha^2 + \alpha^3 = \alpha^2(1 + \alpha) = 0,$$

which is impossible. Therefore μ has order 15.

Multiplication by μ consequently acts transitively on the 15 nonzero elements of K . $\square$

One explicit orientation of the cycle is

$$\begin{aligned} e_0 &= (1, 1, 0, 0, 0), \\ e_1 &= (0, 1, 0, 0, 1), \\ e_2 &= (1, 1, 0, 1, 1), \\ e_3 &= (0, 1, 1, 0, 0), \\ e_4 &= (1, 0, 1, 0, 0), \\ e_5 &= (1, 1, 1, 0, 1), \\ e_6 &= (0, 0, 1, 1, 0), \\ e_7 &= (0, 1, 0, 1, 0), \\ e_8 &= (1, 1, 1, 1, 0), \\ e_9 &= (0, 0, 0, 1, 1), \\ e_{10} &= (0, 0, 1, 0, 1), \\ e_{11} &= (0, 1, 1, 1, 1), \\ e_{12} &= (1, 0, 0, 0, 1), \\ e_{13} &= (1, 0, 0, 1, 0), \\ e_{14} &= (1, 0, 1, 1, 1), \end{aligned}$$

with

$$P(e_q) = e_{q+1}, \quad e_{15} = e_0.$$

4.5.2 Corollary 4.2

For every $X \in \mathbb{R}_+^5$, exactly one of the following holds.

Zero-attractor case

$P^q(X) \longrightarrow 0.$

Nonzero-attractor case There are $c > 0$ and a phase $r \in 0, \dots, 14$ such that

$$\left| P^q(X) - c, e_{q+r \pmod{15}} \right|_{\infty} \longrightarrow 0. \quad (4.2)$$

Thus the possible asymptotic shapes for $n = 5$ are completely classified.

4.6 5. A nonconstant 5-tuple converging to zero

The zero-attractor case is not confined to constant vectors.

Let $\rho \in (0, 1)$ be a root of

$$\rho^4 + 3\rho^3 + 2\rho^2 - 2\rho - 3 = 0. \quad (5.1)$$

Such a root exists because the polynomial is negative at 0 and positive at 1. Numerically,

$$\rho \approx 0.9275619755.$$

Define

$$V = (1, 1 - \rho, 1 - \rho^2, (1 + \rho)(1 - \rho^2), (1 + \rho)^2(1 - \rho^2)).$$

The identity

$$(1 + \rho)^3(1 - \rho^2) - 1 = -\rho(\rho^4 + 3\rho^3 + 2\rho^2 - 2\rho - 3)$$

shows that

$$(1 + \rho)^3(1 - \rho^2) = 1.$$

The coordinates satisfy

$$0 < v_2 < v_3 < v_4 < v_5 < v_1.$$

Moreover,

$$v_1 - v_2 = \rho v_1,$$

$$v_3 - v_2 = \rho v_2,$$

$$v_4 - v_3 = \rho v_3,$$

$$v_5 - v_4 = \rho v_4,$$

$$v_1 - v_5 = \rho v_5.$$

Therefore

$$\boxed{P(V) = \rho V.}$$

Consequently,

$$P^q(V) = \rho^q V \longrightarrow 0,$$

but $P^q(V) \neq 0$ for every finite q .

4.7 6. An exact non-eventually-periodic $n = 5$ orbit with nonzero attractor

This is the more delicate phenomenon. Earlier work explicitly asked whether the $n = 5$ situation permits an orbit which approaches a nontrivial periodic cycle without ever becoming periodic. ([fq.math.ca][2])

The following gives such an orbit exactly.

4.7.1 Theorem 6.1

There exists $X \in \mathbb{R}_+^5$ such that

$$|P^q(X) - e_{q \bmod 15}|_\infty \longrightarrow 0,$$

but no $P^q(X)$ is periodic.

Construction Let

$$p(t) = t^3 - 19t^2 + 139t - 1.$$

We have

$$p\left(\frac{1}{139}\right) = -\frac{2640}{2685619} < 0,$$

and

$$p\left(\frac{1}{138}\right) = \frac{16423}{2628072} > 0.$$

Also $p'(t) > 0$ on this interval. Let

$$\lambda \in \left(\frac{1}{139}, \frac{1}{138}\right)$$

be its unique root. Numerically,

$$\lambda \approx 0.007201330580038611.$$

Define

$$w_0 = \begin{pmatrix} 19960 \\ 83\lambda^2 - 424\lambda + 18633 \\ 22\lambda^2 - 1916\lambda + 6502 \\ 112\lambda^2 - 2496\lambda + 9512 \\ 7\lambda^2 - 156\lambda + 4337 \end{pmatrix}. \quad (6.1)$$

For a sign word $\sigma = (\sigma_1, \dots, \sigma_5) \in +, -^5$, let B_σ be the matrix defined by

$$(B_\sigma v)_i = s_i(v_i - v_{i+1}), \quad s_i = \begin{cases} 1, & \sigma_i = +, \\ -1, & \sigma_i = -, \end{cases}$$

with cyclic indices.

Use the following 15 sign words:

q	σ_q	q	σ_q	q	σ_q
0	+ + - + -	5	+ - + - -	10	+ - + - +
1	- + - - +	6	- - + + -	11	- + - + +
2	+ + - - +	7	- + - + +	12	+ + + - -
3	- - + + +	8	+ + - + -	13	+ - - + -
4	+ - + - -	9	+ - - - +	14	+ - - + -

(6.2)

Define recursively

$$w_{q+1} = B_{\sigma_q} w_q \quad (0 \leq q < 15).$$

The product is

$$A = B_{\sigma_{14}} \cdots B_{\sigma_0} = \begin{pmatrix} 2 & -1 & -15 & 3 & 11 \\ -9 & 10 & -13 & 5 & 7 \\ 11 & -11 & 10 & -7 & -3 \\ 1 & -1 & 9 & -4 & -5 \\ 1 & -1 & -3 & 1 & 2 \end{pmatrix}. \quad (6.3)$$

If $w_0(t)$ denotes the vector in (6.1) with λ replaced by t , direct multiplication gives

$$Aw_0(t) - tw_0(t) = p(t) (083221127). \quad (6.4)$$

Hence

$$w_{15} = Aw_0 = \lambda w_0. \quad (6.5)$$

The finite exact verification following the proof establishes two further facts:

$$w_{q,i} > 0 \quad (0 \leq q < 15, 1 \leq i \leq 5), \quad (6.6)$$

and whenever $e_{q,i} = e_{q,i+1}$, the corresponding sign in σ_q is the sign of

$$w_{q,i} - w_{q,i+1}. \quad (6.7)$$

Let

$$K = \max_{\substack{0 \leq q < 15 \\ 1 \leq i \leq 5}} |w_{q,i} - w_{q,i+1}|, \quad W = \max_{\substack{0 \leq q < 15 \\ 1 \leq i \leq 5}} w_{q,i}.$$

Choose $\varepsilon > 0$ so small that

$$\varepsilon K < \frac{1}{2}, \quad \varepsilon W < \frac{1}{2}.$$

Set

$$X = e_0 + \varepsilon w_0. \quad (6.8)$$

Proof of the claimed dynamics For every q , consider

$$e_q + \delta w_q$$

with $0 < \delta \leq \varepsilon$.

If $e_{q,i} \neq e_{q,i+1}$, their difference is ± 1 , and $\delta K < 1/2$ ensures that the perturbation cannot change its sign.

If $e_{q,i} = e_{q,i+1}$, condition (6.7) determines the sign of the perturbed difference.

Therefore

$$P(e_q + \delta w_q) = e_{q+1} + \delta w_{q+1}. \quad (6.9)$$

Using $w_{15} = \lambda w_0$ and induction gives the exact formula

$$\boxed{P^{15k+q}(X) = e_q + \varepsilon \lambda^k w_q, \quad k \geq 0, \quad 0 \leq q < 15.} \quad (6.10)$$

Since $0 < \lambda < 1$,

$$P^{15k+q}(X) \longrightarrow e_q.$$

Thus the orbit approaches the nonzero 15-cycle.

On the other hand, every coordinate in (6.10) is strictly positive by (6.6). Every nonzero periodic point for $n = 5$ is a scalar multiple of an even-weight binary vector, and hence has at least one zero coordinate. Thus no iterate in (6.10) is periodic.

This proves Theorem 6.1. $\square$

4.7.2 Exact finite verifier

This is not a floating-point experiment. All inequalities are certified using rational interval arithmetic on

$$\lambda \in \left[\frac{1}{139}, \frac{1}{138} \right]$$

and reductions modulo $p(\lambda) = 0$.

```
from fractions import Fraction
import sympy as sp

t = sp.symbols("t")
p = t**3 - 19*t**2 + 139*t - 1
lo, hi = Fraction(1, 139), Fraction(1, 138)

def peval(x):
    return x**3 - 19*x**2 + 139*x - 1

# Existence and uniqueness of the small positive root.
assert peval(lo) < 0 < peval(hi)
assert 3*lo*lo - 38*hi + 139 > 0

cycle = [
    (1,1,0,0,0),
    (0,1,0,0,1),
    (1,1,0,1,1),
    (0,1,1,0,0),
    (1,0,1,0,0),
    (1,1,1,0,1),
    (0,0,1,1,0),
```

```

(0,1,0,1,0),
(1,1,1,1,0),
(0,0,0,1,1),
(0,0,1,0,1),
(0,1,1,1,1),
(1,0,0,0,1),
(1,0,0,1,0),
(1,0,1,1,1),
]

words = [
    "+++-+", "-+---+", "++---+", "---+++",
    "+-+--", "+-+--", "-+---+", "-+---+",
    "+++-+", "+-+---+", "+-+---+", "-+---+",
    "+++-+", "+-+---+", "+-+---+",
]

def branch_matrix(word):
    B = sp.zeros(5)
    for i, symbol in enumerate(word):
        s = 1 if symbol == "+" else -1
        B[i, i] = s
        B[i, (i + 1) % 5] = -s
    return B

def reduce_mod_p(expr):
    return sp.rem(
        sp.Poly(sp.expand(expr), t),
        sp.Poly(p, t),
    ).as_expr()

def reduce_vector(v):
    return sp.Matrix([
        sp.expand(reduce_mod_p(x)) for x in v
    ])

w0 = sp.Matrix([
    19960,
    83*t**2 - 424*t + 18633,
    22*t**2 - 1916*t + 6502,

```

```

        112*t**2 - 2496*t + 9512,
        7*t**2 - 156*t + 4337,
    ])

ws = [w0]
A = sp.eye(5)

for q, word in enumerate(words):
    # Check the binary cycle and fixed signs on unequal edges.
    assert tuple(
        abs(cycle[q][i] - cycle[q][(i + 1) % 5])
        for i in range(5)
    ) == cycle[(q + 1) % 15]

    for i, symbol in enumerate(word):
        if cycle[q][i] != cycle[q][(i + 1) % 5]:
            s = 1 if symbol == "+" else -1
            assert s * (
                cycle[q][i] - cycle[q][(i + 1) % 5]
            ) == 1

    B = branch_matrix(word)
    A = B * A
    ws.append(reduce_vector(B * ws[-1]))

A_expected = sp.Matrix([
    [ 2, -1, -15,  3, 11],
    [-9, 10, -13,  5,  7],
    [11,-11,  10, -7, -3],
    [ 1, -1,  9, -4, -5],
    [ 1, -1, -3,  1,  2],
])

assert A == A_expected
assert all(reduce_mod_p(x) == 0 for x in A*w0 - t*w0)
assert all(reduce_mod_p(x) == 0 for x in ws[15] - t*w0)

def coefficients(expr):
    poly = sp.Poly(expr, t, domain=sp.QQ)
    out = []

```

```

    for degree in range(3):
        c = poly.coeff_monomial(t**degree)
        out.append(Fraction(int(c.p), int(c.q)))
    return out

def interval(expr):
    powers = [
        (Fraction(1), Fraction(1)),
        (lo, hi),
        (lo*lo, hi*hi),
    ]

    lower = Fraction(0)
    upper = Fraction(0)

    for coefficient, (a, b) in zip(coefficients(expr), powers):
        if coefficient >= 0:
            lower += coefficient * a
            upper += coefficient * b
        else:
            lower += coefficient * b
            upper += coefficient * a

    return lower, upper

lower_bounds = []

for q in range(15):
    for i in range(5):
        # Positivity of every perturbation coordinate.
        lower_bounds.append(interval(ws[q][i])[0])

        # Correct sign whenever the binary edge is zero.
        j = (i + 1) % 5
        if cycle[q][i] == cycle[q][j]:
            s = 1 if words[q][i] == "+" else -1
            lower_bounds.append(
                interval(s * (ws[q][i] - ws[q][j]))[0]
            )

```

```

minimum = min(lower_bounds)

print("Return matrix:")
print(A)
print("Certified minimum:", minimum, float(minimum))

assert minimum == Fraction(28286630, 1333149)
assert minimum > 21

```

The certified lower bound is

$$\frac{28286630}{1333149} > 21,$$

so all required inequalities are strict with a wide margin.

4.8 7. Exact algorithm for rational starting vectors

Suppose

$$X = (x_1, \dots, x_n) \in \mathbb{Q}_+^n.$$

Choose a common denominator D and put

$$A = DX \in \mathbb{Z}_+^n.$$

Positive homogeneity gives

$$P^q(X) = \frac{1}{D} P^q(A).$$

Let

$$R = \max_i A_i - \min_i A_i.$$

Every coordinate of $P(A)$ lies in $0, \dots, R$, and the same is true for every later iterate. Thus after the first step there are at most

$$(R + 1)^n$$

possible states.

The following exact algorithm therefore terminates.

INPUT: rational vector X of length n

1. Let D be the least common multiple of all denominators.
2. Set $A := D X$, an integer vector.
3. Set $Y := P(A)$.
4. Initialize an empty dictionary `seen`.
5. For $q = 1, 2, 3, \dots$:
 - if Y is already in `seen`:
 - $\mu := \text{seen}[Y]$
 - $\square := q - \mu$
 - return transient length μ ,
 - cycle length $\square$,
 - exact cycle divided by D
 - `seen`[Y] := q
 - $Y := P(Y)$

Each iteration takes $O(n)$ integer operations on integers having $O(\log R)$ bits.

The crude worst-case bounds are therefore

$$O(n(R+1)^n)$$

arithmetic operations and

$$O(n(R+1)^n \log R)$$

bit storage. Floyd's cycle-finding method reduces the memory to $O(n \log R)$ bits, at the cost of rerunning part of the orbit to output the cycle.

The same procedure applies whenever there exist $a \neq 0$ and $b \in \mathbb{R}$ such that

$$a(X + b\mathbf{1}) \in \mathbb{Q}^n,$$

because

$$P(X + b\mathbf{1}) = P(X), \quad P(aX) = |a|P(X).$$

4.8.1 Consequence for rational 5-tuples

For odd n , no positive constant vector lies in the image of P . Indeed, if

$$P(Y) = c\mathbf{1}, \quad c > 0,$$

then there are signs $s_i \in \pm 1$ such that

$$y_i - y_{i+1} = s_i c.$$

Summing cyclically gives

$$0 = c \sum_{i=1}^n s_i,$$

which is impossible when n is odd.

It follows that for odd n , only constant starting vectors can ever reach zero exactly.

Therefore, for $n = 5$:

$X \in \mathbb{Q}_+^5, X \text{ nonconstant} \implies \text{the orbit eventually enters the exact period-15 cycle.}$
--

4.9 8. Small and boundary cases

Constant vectors If $X = c\mathbf{1}$, then $P(X) = 0$.

$n = 3$ Here $h = 1$, so $\mathcal{S}_3$ consists of the even-weight binary vectors:

$$000, \quad 110, \quad 011, \quad 101.$$

The three nonzero vectors form the cycle

$$110 \mapsto 011 \mapsto 101 \mapsto 110.$$

Every real 3-orbit is asymptotic either to 0 or to a scalar multiple of this 3-cycle.

$n = 4$ Here $h = 4$, so

$$\mathcal{S}_4 = 0000.$$

Thus every real orbit converges to zero. This does not imply finite extinction: the algebraic exceptional family mentioned in the editorial note converges to zero without reaching it.

$n = 5$ There is one nonzero 15-cycle up to scale and phase. Both nonperiodic possibilities occur:

- the eigenray of Section 5 approaches zero;
- the construction of Section 6 approaches the nonzero 15-cycle.

Necessity of the power-of-two hypothesis If n is not a power of 2, then

$$|\mathcal{S}_n| = 2^{n-h} > 1,$$

so nonzero periodic orbits exist. Hence “every orbit converges to zero” is equivalent to n being a power of 2.

Nonnegativity The restriction to $\mathbb{R}_+^n$ is not essential: P sends every vector in $\mathbb{R}^n$ into $\mathbb{R}_+^n$ after one iteration.

4.10 9. Exact remaining obstruction

The asymptotic shape is now determined for every n , and completely explicit for $n = 5$. What is not supplied by Theorem 2.1 is a finite intrinsic criterion for an arbitrary real input X deciding whether

$$c(X) = \lim_{q \rightarrow \infty} |P^q(X)|_\infty$$

is zero or positive, nor a closed formula for $c(X)$ and the eventual phase. For $n = 5$, the exact obstruction can be formulated as follows.

Near the 15-cycle, the 15-step return map is piecewise linear. Each branch is a product of 15 signed difference matrices. A complete basin classification would require describing all admissible infinite paths through the resulting finite polyhedral branch graph and determining which associated matrix products contract the perturbation while preserving a positive limiting scale.

The construction in Section 6 gives one exact contracting return branch. A natural next lemma would be:

Proposed next lemma. Construct the finite directed graph of admissible polyhedral cones for the 15-step return map at e_0 , and classify all strongly connected components whose matrix cocycle has a nontrivial stable direction.

That would turn the remaining $n = 5$ basin problem into a finite exact computation followed by a matrix-semigroup argument.

Thus the strongest justified status is:

Complete classification of all attractors for every n ;
complete asymptotic classification for $n = 5$;
exact rational-input algorithm;
explicit nonperiodic $n = 5$ examples for both $c = 0$ and $c > 0$;
general real basin classification still remaining.

[1]: <https://hrj.episciences.org/105/pdf> "https://hrj.episciences.org/105/pdf"

[2]: https://www.fq.math.ca/Papers1/45-2/quartzerr02_2007.pdf "https://www.fq.math.ca/Papers1/45-2/quartzerr02_2007.pdf"

Formal verification. Lean source; online type-check.

5 Question Q728

Informal verdict. A — **Complete solution**, with high confidence, for the meaningful range $n \geq 3$.

Lean coverage. The Lean development proves the control and steering lemmas and the complete minimax outcome theorem for every n at least 3. The lower bound is the necessary domain repair because the printed last-mover payoff is undefined for $n=1,2$.

5.1 Audited informal answer

5.1.1 1. Faithful restatement and a domain issue

A position is a finite collection of sticks of positive integer lengths. A legal move replaces one stick of length m by two sticks of distinct positive integer lengths $a < b$ satisfying $a + b = m$. Hence a stick is splittable exactly when $m \geq 3$.

Player A makes the first move, then B , and they alternate until every remaining stick has length 1 or 2. If the terminal position contains x sticks of length 1 and y sticks of length 2, then:

- if $x > y$, the last mover wins;
- if $x < y$, the last mover loses;

- if $x = y$, the game is drawn.

The question asks whether, under optimal play, the outcome depends only on $n \bmod 3$ in the stated way. This is the complete statement of Q728 in the canonical PDF.

There is one genuine omission: the PDF does not explicitly assume $n \geq 3$. For $n = 1, 2$, A has no legal first move and there is no “last player”, so the payoff rule is undefined. I therefore solve the intended problem for $n \geq 3$.

Theorem. For every integer $n \geq 3$:

$$\begin{cases} A \text{ wins,} & n \equiv 2 \pmod{3}, \\ B \text{ wins,} & n \equiv 1 \pmod{3}, \\ \text{the game is drawn,} & n \equiv 0 \pmod{3}. \end{cases}$$

Thus the proposed assertion is true after inserting the necessary hypothesis $n \geq 3$.

The proof is constructive and gives explicit optimal strategies.

5.1.2 2. Elementary terminal identities

Let a terminal position contain x pieces of length 1 and y pieces of length 2. Conservation of total length gives

$$x + 2y = n.$$

Consequently,

$$x + y = n - y.$$

After A ’s opening move there are exactly two pieces, and B is to move. Since every subsequent move increases the number of pieces by one, the number of moves remaining after the opening is

$$R = (x + y) - 2 = n - y - 2.$$

Therefore:

- B is the overall last mover exactly when R is odd;
- A is the overall last mover exactly when R is even, including $R = 0$, since A then made the opening and final move.

The main task is thus to control the possible terminal value of y .

5.2 3. The control interval of a position

For a positive integer m , define

$$\beta(m) = \begin{cases} 0, & m = 1, \\ 1, & m = 2, \\ \lfloor m/3 \rfloor, & m \geq 3, \end{cases}$$

and

$$\eta(m) = \begin{cases} 1, & m \geq 5 \text{ and } m \equiv 2 \pmod{3}, \\ 0, & \text{otherwise.} \end{cases}$$

We call a piece counted by η a **hot piece**.

For a position $P = (m_1, \dots, m_k)$, put

$$b(P) = \sum_{i=1}^k \beta(m_i), \quad h(P) = \sum_{i=1}^k \eta(m_i),$$

and define its **control interval**

$$J(P) = \left\{ b(P) + \left\lfloor \frac{h(P)}{2} \right\rfloor, b(P) + \left\lceil \frac{h(P)}{2} \right\rceil \right\}.$$

Thus $J(P)$ is either a singleton or a pair of consecutive integers.

If P is terminal, every piece is 1 or 2. Hence $h(P) = 0$, while $b(P)$ is precisely the number y of pieces of length 2. Therefore

$$\boxed{P \text{ terminal} \implies J(P) = y.}$$

This is why $J(P)$ is useful.

5.2.1 4. Effect of one move

Suppose a piece m is split into $a + b = m$, with $1 \leq a < b$. Write

$$\Delta b = \beta(a) + \beta(b) - \beta(m), \quad \Delta h = \eta(a) + \eta(b) - \eta(m).$$

The following table exhausts all possibilities. In the table, residue classes of the children are unordered.

$m \bmod 3$	Child residues	Additional condition	$(\Delta b, \Delta h)$
0	0 + 0	—	$((0,0))$
0	1 + 2	the 2 mod 3 child is 2	$((0,0))$
0	1 + 2	the 2 mod 3 child is at least 5	$((-1,1))$
1	0 + 1	—	$((0,0))$
1	2 + 2	exactly one child is 2	$((0,1))$
1	2 + 2	both children are at least 5	$((-1,2))$
2	1 + 1	—	$((0,-1))$
2	0 + 2	the 2 mod 3 child is 2	$((1,-1))$
2	0 + 2	the 2 mod 3 child is at least 5	$((0,0))$

Here a splittable piece congruent to 2 mod 3 is necessarily at least 5, so it is itself hot.

For completeness, the table follows immediately from

$$\begin{aligned}
(\beta(3r), \eta(3r)) &= (r, 0), \\
(\beta(3r + 1), \eta(3r + 1)) &= (r, 0), \\
(\beta(2), \eta(2)) &= (1, 0), \\
(\beta(3r + 2), \eta(3r + 2)) &= (r, 1), \quad r \geq 1,
\end{aligned}$$

together with the possible residue decompositions

$$\begin{aligned}
0 &= 0 + 0 = 1 + 2, \\
1 &= 0 + 1 = 2 + 2, \quad (\bmod 3). \\
2 &= 0 + 2 = 1 + 1
\end{aligned}$$

Define the half-integral centre

$$c(P) = b(P) + \frac{h(P)}{2}.$$

The table shows that under every move $P \rightarrow P'$,

$$c(P') - c(P) \in \left\{ -\frac{1}{2}, 0, \frac{1}{2} \right\}.$$

Since

$$J(P) = \lfloor c(P) \rfloor, \lceil c(P) \rceil,$$

we obtain the following overlap property.

Lemma 1 — Overlap under every move For every legal move $P \rightarrow P'$,

$$\boxed{J(P) \cap J(P') \neq \emptyset.}$$

Indeed, two half-integers differing by at most $(1/2)$ have integer hulls sharing at least one integer.

5.3 5. The steering lemma

Fix two consecutive integers

$$T_t = t, t + 1.$$

Lemma 2 — One-move steering Suppose P is nonterminal and

$$J(P) \cap T_t \neq \emptyset.$$

Then there is a legal move $P \rightarrow P'$ such that

$$\boxed{J(P') \subseteq T_t.}$$

Proof Write $b = b(P)$ and $h = h(P)$.

Case 1: h is odd Write $h = 2r + 1$. Then

$$J(P) = b + r, b + r + 1.$$

Because $h > 0$, there is a hot piece $m \equiv 2 \pmod{3}$, $m \geq 5$. Both of the following splits are legal:

$$m = 1 + (m - 1), \quad m = 2 + (m - 2).$$

From the move table:

- the first split has $(\Delta b, \Delta h) = (0, -1)$, and therefore

$$J(P') = b + r;$$

- the second has $(\Delta b, \Delta h) = (1, -1)$, and therefore

$$J(P') = b + r + 1.$$

At least one of the two endpoints of $J(P)$ belongs to T_t . Choose the corresponding split. The new interval is then a singleton contained in T_t .

Case 2: $h > 0$ is even Write $h = 2r$. Then

$$J(P) = s, \quad s = b + r.$$

Since $J(P) \cap T_t \neq \emptyset$, either $s = t$ or $s = t + 1$.
Again choose a hot piece m .

- Splitting $m = 1 + (m - 1)$ gives

$$J(P') = s - 1, s.$$

- Splitting $m = 2 + (m - 2)$ gives

$$J(P') = s, s + 1.$$

Thus:

- if $s = t$, use $(2 + m - 2)$;
- if $s = t + 1$, use $(1 + m - 1)$.

In either case $J(P') = T_t$.

Case 3: $h = 0$ Then $J(P) = b$, and $b \in T_t$.

Because P is nonterminal, it contains a splittable piece $m \geq 3$. Since $h = 0$, no splittable piece can be congruent to $2 \pmod 3$. Hence $m \equiv 0$ or $1 \pmod 3$.

- If $m = 3$, split it as $1 + 2$.
- If $m \equiv 0 \pmod 3$ and $m \geq 6$, split it as

$$m = 2 + (m - 2).$$

- If $m \equiv 1 \pmod 3$, split it as

$$m = 1 + (m - 1).$$

Each of these moves has $(\Delta b, \Delta h) = (0, 0)$, so

$$J(P') = J(P) = b \subseteq T_t.$$

This proves the lemma. $\square$

5.4 6. The interval-control theorem

The previous two lemmas combine into a robust strategy.

Theorem 3 — A designated player can control the terminal value

Let P_0 be a position satisfying

$$J(P_0) \subseteq T_t = t, t + 1.$$

Designate either of the two players as the **controller**. It does not matter whether the controller or the opponent is next to move.

Then the controller has a strategy forcing the terminal number y of length-2 pieces to satisfy

$$\boxed{y \in t, t + 1.}$$

Proof The controller maintains the following invariant:

- immediately after every controller move,

$$J(P) \subseteq T_t;$$

- immediately after every opponent move,

$$J(P) \cap T_t \neq \emptyset.$$

Initially $J(P_0) \subseteq T_t$.

Suppose the controller has just moved and $J(P) \subseteq T_t$. If the opponent makes a move $P \rightarrow Q$, Lemma 1 gives

$$J(P) \cap J(Q) \neq \emptyset.$$

Since $J(P) \subseteq T_t$, it follows that

$$J(Q) \cap T_t \neq \emptyset.$$

If Q is terminal, then $J(Q) = y$, so $y \in T_t$. If Q is nonterminal, Lemma 2 gives the controller a move $Q \rightarrow Q'$ with

$$J(Q') \subseteq T_t.$$

The same argument applies when the opponent is initially first to move.

Finally, the game necessarily terminates: every move increases the number of pieces by one, while the number of pieces is at most n . $\square$

This theorem is stronger than what is needed: either player can be designated as controller, independently of whose turn it is.

5.5 7. Control intervals for two initial pieces

Suppose the current position consists of two pieces whose total length is N , and write

$$N = 3q + r, \quad r \in 0, 1, 2.$$

For residue bookkeeping, distinguish:

- type 0: length $3i$;
- type 1: length $3i + 1$;
- type 2_0 : the exceptional length 2;
- type 2_1 : length $3i + 2 \geq 5$.

Direct substitution in the definitions of β and η gives:

$N \bmod 3$	Types of the two pieces	$J(P)$
0	$0 + 0$	$(\{q\})$
0	$1 + 2_0$	$(\{q\})$
0	$1 + 2_1$	$(\{q-1, q\})$
1	$0 + 1$	$(\{q\})$
1	$2_0 + 2_0$	$(\{q+1\})$
1	$2_0 + 2_1$	$(\{q, q+1\})$
1	$2_1 + 2_1$	$(\{q\})$
2	$1 + 1$	$(\{q\})$
2	$0 + 2_0$	$(\{q+1\})$
2	$0 + 2_1$	$(\{q, q+1\})$

For example, in the last row the pieces are $3i$ and $3j + 2$, with $j \geq 1$. Then $q = i + j$,

$$b(P) = i + j = q, \quad h(P) = 1,$$

so $J(P) = q, q + 1$. The other rows follow identically.

We therefore have the two key inclusions:

$$\boxed{N \equiv 1, 2 \pmod{3} \implies J(P) \subseteq q, q + 1,}$$

and

$$\boxed{N \equiv 0 \pmod{3} \implies J(P) \subseteq q-1, q.}$$

These statements hold for every two-piece position of total length N , whether or not the two lengths are distinct.

5.6 8. Determination of the winner

After A 's opening move, there are two pieces and B is next.

5.6.1 Case 1: $n = 3q + 1$

Every opening produces a two-piece position P satisfying

$$J(P) \subseteq q, q+1.$$

Designate B as controller. By Theorem 3, B can force

$$y \in q, q+1.$$

We check both possibilities.

If $y = q$ Then

$$x = n - 2y = 3q + 1 - 2q = q + 1 > q = y.$$

Also,

$$R = n - y - 2 = 3q + 1 - q - 2 = 2q - 1$$

is odd, so B is the last mover. Since $x > y$, the last mover wins. Thus B wins.

If $y = q + 1$ Then

$$x = 3q + 1 - 2(q + 1) = q - 1 < q + 1 = y.$$

Moreover,

$$R = 3q + 1 - (q + 1) - 2 = 2q - 2$$

is even, so A is the last mover. Since $x < y$, the last mover loses. Thus B again wins.

Therefore

$$\boxed{n \equiv 1 \pmod{3} \implies B \text{ wins.}}$$

5.6.2 Case 2: $n = 3q + 2$

Here $n \geq 5$, since $n = 2$ lies outside the legal range.

Every opening gives a position satisfying

$$J(P) \subseteq q, q + 1.$$

This time designate A as controller. Although B is next to move, Theorem 3 applies regardless of who moves next. Hence A can force

$$y \in q, q + 1.$$

If $y = q$ Then

$$x = 3q + 2 - 2q = q + 2 > q = y,$$

and

$$R = 3q + 2 - q - 2 = 2q$$

is even. Therefore A is the last mover and wins.

If $y = q + 1$ Then

$$x = 3q + 2 - 2(q + 1) = q < q + 1 = y,$$

while

$$R = 3q + 2 - (q + 1) - 2 = 2q - 1$$

is odd. Thus B is the last mover, but because $x < y$, the last mover loses. Hence A wins.

Therefore

$$\boxed{n \equiv 2 \pmod{3} \implies A \text{ wins.}}$$

In fact, after any legal opening, A can use the interval-control strategy.

5.6.3 Case 3: $n = 3q$

We prove separately that B cannot lose and that A cannot lose.

B has a non-losing strategy Whatever opening A chooses, the resulting two-piece position satisfies

$$J(P) \subseteq q - 1, q.$$

Designate B as controller. Then B can force

$$y \in q - 1, q.$$

- If $y = q$, then $x = q$, so the game is drawn.
- If $y = q - 1$, then

$$x = 3q - 2(q - 1) = q + 2 > q - 1 = y,$$

and

$$R = 3q - (q - 1) - 2 = 2q - 1$$

is odd. Hence B is the last mover and wins.

Thus B can force either a win or a draw.

A has a non-losing strategy Player A chooses the following opening:

$$\begin{cases} 1 + 2, & n = 3, \\ 2 + 4, & n = 6, \\ 3 + (n - 3), & n \geq 9. \end{cases}$$

All these splits are legal and have distinct positive parts. Their control interval is exactly

$$J(P) = q.$$

Indeed:

- for $n = 3$, $b(1) + b(2) = 1 = q$;
- for $n = 6$, $b(2) + b(4) = 1 + 1 = 2 = q$;

- for $n = 3q \geq 9$,

$$b(3) + b(3q - 3) = 1 + (q - 1) = q,$$

and there are no hot pieces.

Now designate A as controller with target

$$q, q + 1.$$

Although B moves next, Theorem 3 allows A to force

$$y \in q, q + 1.$$

- If $y = q$, the game is drawn.
- If $y = q + 1$, then, whenever this value is feasible,

$$x = 3q - 2(q + 1) = q - 2 < q + 1 = y,$$

and

$$R = 3q - (q + 1) - 2 = 2q - 3$$

is odd. Thus B is the last mover and loses, so A wins.

Hence A can force either a win or a draw.

Since both players possess non-losing strategies, neither can force a win against optimal opposition. Therefore

$$\boxed{n \equiv 0 \pmod{3} \implies \text{draw.}}$$

If both players use the displayed control strategies, B forces $y \in q - 1, q$ while A forces $y \in q, q + 1$; consequently $y = q$, so the resulting play is necessarily a draw.

5.7 9. Explicit optimal strategy and complexity

The proof gives a direct algorithm.

Suppose a player has been designated as controller for a target

$$T_t = t, t + 1.$$

At each controller turn, compute $b(P)$, $h(P)$, and $J(P)$.

When $h(P)$ is odd Choose any hot piece m .

- Splitting $m = 1 + (m - 1)$ makes the new interval the lower endpoint of $J(P)$.
- Splitting $m = 2 + (m - 2)$ makes it the upper endpoint.

Choose an endpoint lying in T_t .

When $h(P) > 0$ is even Then $J(P) = s$, where $s = t$ or $s = t + 1$. Choose a hot piece m .

- If $s = t$, use $m = 2 + (m - 2)$.
- If $s = t + 1$, use $m = 1 + (m - 1)$.

The resulting interval is exactly T_t .

When $h(P) = 0$ Choose any splittable piece m .

- If $m = 3$, play $1 + 2$.
- If $m \equiv 0 \pmod{3}$ and $m \geq 6$, play $(2+m-2)$.
- If $m \equiv 1 \pmod{3}$, play $(1+m-1)$.

This preserves $J(P)$.

The appropriate controllers and targets are:

$n \bmod 3$	controller after the opening	target
1	B	$q, q + 1$
2	A	$q, q + 1$
0	B	$q - 1, q$

For $n \equiv 0 \pmod{3}$, A additionally uses the special opening above and controls $(\{q, q+1\})$.

If the multiset of pieces is maintained together with $b(P)$, $h(P)$, and buckets for hot and non-hot active pieces, every update and every strategic choice takes $O(1)$ word operations. There are at most $n - 1$ moves, since every move increases the number of pieces and there can be at most n pieces. Thus the full strategy is implementable in

$$O(n)$$

word operations and $O(n)$ storage. A naive implementation that scans all current pieces at every turn has $O(n^2)$ worst-case running time.

5.8 10. Boundary checks

$n = 3$ The only move is

$$3 \longrightarrow 1 + 2.$$

Thus $x = y = 1$, and the game is drawn.

$n = 4$ The only opening is

$$4 \longrightarrow 1 + 3.$$

Then B must split $3 \rightarrow 1 + 2$. The terminal position has $x = 2, y = 1$, and B is the last mover, so B wins.

$n = 5$ Player A may play

$$5 \longrightarrow 2 + 3.$$

Player B must split $3 \rightarrow 1 + 2$. The terminal pieces are $(1, 2, 2)$, so $x = 1 < 2 = y$. Since B is the last mover, B loses and A wins.

$n = 6$ Player A uses

$$6 \longrightarrow 2 + 4.$$

Then the continuation is forced:

$$4 \longrightarrow 1 + 3, \quad 3 \longrightarrow 1 + 2.$$

The terminal position has $x = y = 2$, hence a draw.

These are exactly the first four cases predicted by the theorem.

5.9 11. Necessity of the hypotheses

- The restriction $n \geq 3$ is necessary merely to make the stated payoff rule well-defined: for $n = 1, 2$, no one ever plays, so there is no last player.
- The requirement that the two new lengths be **distinct** is essential. If equal splits were permitted, then for $n = 3$ one would have

$$3 \rightarrow 1 + 2 \rightarrow 1 + 1 + 1,$$

so B , rather than producing a draw, would be the last mover with $x > y$ and would win.

- Integrality is used both for termination at pieces of lengths (1,2) and for the residue-3 control invariant.

5.10 12. Exact computational verification

The proof above is independent of computation. The following exact dynamic program was used only as a discovery and consistency check. It verifies the claimed pattern for $3 \leq n \leq 40$.

```
from functools import cache

def legal_moves(position: tuple[int, ...]) -> set[tuple[int, ...]]:
    """All legal successor positions."""
    successors: set[tuple[int, ...]] = set()

    for i, m in enumerate(position):
        # a < m-a, so equal splits are automatically excluded.
        for a in range(1, (m + 1) // 2):
            b = m - a
            if a < b:
                nxt = list(position)
                nxt.pop(i)
                nxt.extend((a, b))
                successors.add(tuple(sorted(nxt)))

    return successors

@cache
def value(position: tuple[int, ...]) -> int:
    """
    Minimax value for the player whose turn it is:
    +1 = that player can force a win,
    0 = that player can force a draw,
```

```

    -1 = that player must lose under optimal play.
    """
    successors = legal_moves(position)

    if not successors:
        x = position.count(1)
        y = position.count(2)

        # At a genuine terminal position, the player to move is
        # the opponent of the last mover. That player wins iff x < y.
        return (y > x) - (y < x)

    # Roles are reversed after every move.
    return max(-value(nxt) for nxt in successors)

expected = {
    0: 0,      # draw
    1: -1,     # A loses, hence B wins
    2: 1,      # A wins
}

for n in range(3, 41):
    assert value((n,)) == expected[n % 3]

print("Verified for every n from 3 through 40.")

```

For a fixed n , the cached verifier visits at most the $p(n)$ integer partitions of n , where p is the partition function. There are $O(n)$ potential splits per state, so the conceptual transition count is $(O(n, pn))$. This is only a finite verification method; the invariant proof establishes the result for all $n \geq 3$.

5.10.1 Final conclusion

The answer to Q728 is **yes**, with the necessary clarification that the game is defined for $n \geq 3$:

$n \equiv 2 \pmod{3} \implies A \text{ wins,}$
$n \equiv 1 \pmod{3} \implies B \text{ wins,}$
$n \equiv 0 \pmod{3} \implies \text{draw.}$

Formal verification. Lean source; online type-check.

6 Question Q756

Informal verdict. A — **Complete solution**, with high confidence.

Lean coverage (partial). Lean proves automatic smoothness and the complete classification of polynomial solutions. The informal answer’s shell construction of an infinite-dimensional family of nonpolynomial solutions is not formalized.

6.1 Audited informal answer

6.1.1 1. Faithful restatement

Let $\beta, \lambda \in \mathbb{R}$, with $|\lambda| > 1$, and define

$$F(\beta, \lambda) = \{g \in C^1(\mathbb{R}, \mathbb{R}) : g'(t) = g(\lambda t) - \beta g(t) \text{ for every } t \in \mathbb{R}\}.$$

The source observes that, for every nonnegative integer p , there is a nonzero polynomial of degree p in $F(\lambda^p, \lambda)$, and asks, for every admissible pair (β, λ) , whether $F(\beta, \lambda)$ contains nonpolynomial functions. There are no additional subparts or footnotes attached to Q756.

There is no substantive typo or ambiguity. The only conventional point is whether “natural integer” includes 0; below,

$$\mathbb{N}_0 = 0, 1, 2, \dots,$$

and the case $p = 0$ is the constant-polynomial case $\beta = 1$.

6.1.2 2. Final theorem

Theorem For every $\beta, \lambda \in \mathbb{R}$ satisfying $|\lambda| > 1$, the space $F(\beta, \lambda)$ contains an infinite-dimensional vector subspace of nonzero C^∞ functions that are flat at the origin:

$$g^{(n)}(0) = 0 \quad (n \geq 0).$$

Consequently,

For every $\beta \in \mathbb{R}$ and every $|\lambda| > 1$, there exist nonpolynomial functions in $F(\beta, \lambda)$.

Moreover, the polynomial solutions are completely classified:

- if $\beta \neq \lambda^p$ for every $p \in \mathbb{N}_0$, the zero function is the only polynomial solution;
- if $\beta = \lambda^p$, the polynomial solution space is one-dimensional and is generated by

$$P_p(t) = \sum_{n=0}^p \frac{1}{n!} \left(\prod_{k=0}^{n-1} (\lambda^k - \lambda^p) \right) t^n,$$

where the empty product, for $n = 0$, equals 1.

The constructed nonpolynomial solutions require no boundedness or growth assumption at infinity.

6.2 3. Automatic regularity

We first record that the C^1 assumption automatically improves to C^∞ .

Lemma 1 Every $g \in F(\beta, \lambda)$ is C^∞ , and for every $n \geq 0$,

$$g^{(n+1)}(t) = \lambda^n g^{(n)}(\lambda t) - \beta g^{(n)}(t). \quad (3.1)$$

In particular,

$$g^{(n+1)}(0) = (\lambda^n - \beta)g^{(n)}(0). \quad (3.2)$$

Proof Suppose $g \in C^m$, with $m \geq 1$. Then

$$t \mapsto g(\lambda t) - \beta g(t)$$

belongs to C^m . Since this function equals (g') , it follows that $g' \in C^m$, hence $g \in C^{m+1}$. Induction gives $g \in C^\infty$.

Differentiating the defining equation n times gives

$$g^{(n+1)}(t) = \frac{d^n}{dt^n} g(\lambda t) - \beta g^{(n)}(t) \lambda^n g^{(n)}(\lambda t) - \beta g^{(n)}(t).$$

Putting $t = 0$ proves (3.2). $\square$

A useful consequence is immediate.

Corollary 2 If $g \in F(\beta, \lambda)$ and $g(0) = 0$, then g is flat at the origin:

$$g^{(n)}(0) = 0 \quad (n \geq 0).$$

Proof Starting with $g(0) = 0$, recurrence (3.2) gives successively

$$g'(0) = g''(0) = \cdots = 0.$$

□

Thus it will suffice to construct nonzero global solutions whose two one-sided limits at 0 are both zero.

6.3 4. Polynomial solutions

Let

$$P(t) = \sum_{k=0}^m a_k t^k, \quad a_m \neq 0,$$

be a polynomial solution of degree m .

The coefficient of t^m in (P') is zero, whereas the coefficient of t^m in

$$P(\lambda t) - \beta P(t)$$

is

$$(\lambda^m - \beta)a_m.$$

Therefore

$$\beta = \lambda^m. \tag{4.1}$$

More generally, comparison of the coefficient of t^k gives

$$(k+1)a_{k+1} = (\lambda^k - \beta)a_k, \quad k \geq 0. \tag{4.2}$$

If $a_0 = 0$, recurrence (4.2) gives $a_1 = a_2 = \cdots = 0$, so every nonzero polynomial solution has $a_0 \neq 0$.

Suppose now that $\beta = \lambda^p$. Taking $a_0 = 1$, recurrence (4.2) yields

$$a_n = \frac{1}{n!} \prod_{k=0}^{n-1} (\lambda^k - \lambda^p).$$

For $n \leq p$, none of the factors vanishes, because $|\lambda| > 1$ implies that the numbers $\lambda^0, \lambda^1, \dots$ are pairwise distinct. At the next step,

$$a_{p+1} = \frac{\lambda^p - \lambda^p}{p+1} a_p = 0,$$

and all subsequent coefficients vanish. This gives the polynomial P_p stated in the theorem.

Conversely, the preceding argument shows that any nonzero polynomial solution of degree m necessarily has $\beta = \lambda^m$. Thus the polynomial classification is complete. □

6.4 5. Extension from a fundamental annulus

Set

$$\rho := |\lambda| > 1$$

and, for $n \in \mathbb{Z}$, define the closed annuli

$$A_n = \{t \in \mathbb{R} : \rho^n \leq |t| \leq \rho^{n+1}\}.$$

The fundamental annulus is

$$A_0 = t : 1 \leq |t| \leq \rho.$$

Let

$$\mathcal{S} = C_c^\infty(t : 1 < |t| < \rho).$$

Thus every $\varphi \in \mathcal{S}$ vanishes on neighborhoods of the four boundary points

$$-\rho, \quad -1, \quad 1, \quad \rho.$$

We show that any such seed φ can be extended uniquely to a smooth solution on $\mathbb{R} \setminus 0$.

6.4.1 5.1. Extension toward infinity

Set

$$G_\varphi(t) = \varphi(t), \quad t \in A_0.$$

Assume that G_φ has been defined on A_n , for some $n \geq 0$. For $x \in A_{n+1}$, define

$$G_\varphi(x) = G'_\varphi! \left(\frac{x}{\lambda} \right) + \beta G_\varphi \left(\frac{x}{\lambda} \right). \quad (5.1)$$

Indeed, $x/\lambda \in A_n$.

By construction,

$$G_\varphi(\lambda t) = G'_\varphi(t) + \beta G_\varphi(t)$$

on A_n , which is exactly the desired equation.

Furthermore,

$$\text{supp}(G_\varphi|_{A_{n+1}}) \subset \lambda, \text{supp}(G_\varphi|_{A_n}). \quad (5.2)$$

Differentiation does not enlarge support. Since the seed is supported strictly inside A_0 , induction shows that every outward piece is supported strictly inside its annulus. Hence adjacent pieces vanish on neighborhoods of their common boundary and glue C^∞ -smoothly.

This defines G_φ on $|t| \geq 1$.

6.4.2 5.2. Extension toward the origin

Suppose that G_φ has already been defined for

$$|t| \geq \rho^{-m},$$

where $m \geq 0$. We define it on the next inner shell

$$A_{-(m+1)} = \left\{ t : \rho^{-(m+1)} \leq |t| \leq \rho^{-m} \right\}.$$

Fix a sign $\sigma \in -1, 1$, and put

$$a = \sigma \rho^{-m}.$$

For t in the corresponding component of $A_{-(m+1)}$, define

$$G_\varphi(t) = e^{-\beta(t-a)} G_\varphi(a) + \int_a^t e^{-\beta(t-s)} G_\varphi(\lambda s) ds. \quad (5.3)$$

For such s ,

$$\rho^{-m} \leq |\lambda s| \leq \rho^{-m+1},$$

so $G_\varphi(\lambda s)$ is already known.

Differentiating (5.3) gives

$$G'_\varphi(t) = -\beta G_\varphi(t) + G_\varphi(\lambda t),$$

as required.

At $t = a$, formula (5.3) reproduces the previously assigned value $G_\varphi(a)$. The old and new pieces solve the same first-order inhomogeneous differential equation near the boundary, with the same boundary value. Since the forcing

$$t \longmapsto G_\varphi(\lambda t)$$

is already C^∞ there, the pieces glue C^∞ -smoothly.

At the initial boundaries $t = \pm 1$, the seed and the forcing both vanish in a neighborhood, so the same argument applies.

Induction defines a function

$$G_\varphi \in C^\infty(\mathbb{R} \setminus 0)$$

satisfying

$$G'_\varphi(t) = G_\varphi(\lambda t) - \beta G_\varphi(t), \quad t \neq 0. \quad (5.4)$$

All operations in the construction are linear in φ .

6.5 6. Uniform boundedness near the origin

We next prove that every annular extension has finite one-sided limits at 0.

For $n \geq 0$, set

$$M_n = \sup_{t \in A_{-n}} |G_\varphi(t)|.$$

Thus $M_0 = |\varphi|_\infty$.

The width of either connected component of A_{-n} , for $n \geq 1$, is

$$\ell_n = \rho^{-n+1} - \rho^{-n}(\rho - 1)\rho^{-n}.$$

Let $t \in A_{-n}$ and let a denote the outer endpoint of its connected component. From (5.3),

$$|G_\varphi(t)| \leq e^{|\beta|\ell_n} |G_\varphi(a)| + e^{|\beta|\ell_n} \int_a^t |G_\varphi(\lambda s)| |ds|.$$

Both $G_\varphi(a)$ and $G_\varphi(\lambda s)$ are bounded in absolute value by M_{n-1} . Hence

$$M_n \leq e^{|\beta|\ell_n} (1 + \ell_n) M_{n-1} \leq e^{(|\beta|+1)\ell_n} M_{n-1}. \quad (6.1)$$

But

$$\sum_{n=1}^{\infty} \ell_n = \sum_{n=1}^{\infty} (\rho^{-n+1} - \rho^{-n}) = 1.$$

Iterating (6.1) gives

$$M_n \leq e^{(|\beta|+1)n} M_0 \quad (n \geq 1). \quad (6.2)$$

Thus G_φ is uniformly bounded on a punctured neighborhood of 0. Since $G_\varphi(\lambda t)$ is also bounded there, equation (5.4) gives a uniform derivative bound:

$$|G'_\varphi(t)| \leq |G_\varphi(\lambda t)| + |\beta| |G_\varphi(t)| \leq C_{\beta, \lambda, \varphi}, \quad 0 < |t| \leq 1. \quad (6.3)$$

Consequently G_φ is uniformly Lipschitz on each side of the origin. Therefore the finite limits

$$L_+(\varphi) = \lim_{t \downarrow 0} G_\varphi(t), \quad L_-(\varphi) = \lim_{t \uparrow 0} G_\varphi(t) \quad (6.4)$$

exist.

The map

$$\Lambda : \mathcal{S} \longrightarrow \mathbb{R}^2, \quad \Lambda(\varphi) = (L_-(\varphi), L_+(\varphi)), \quad (6.5)$$

is linear.

6.6 7. Killing both one-sided limits

The seed space $\mathcal{S}$ is infinite-dimensional. More concretely, inside the nonempty interval $(1, \rho)$ one may choose arbitrarily many pairwise disjoint intervals and place a nonzero smooth bump function in each one.

For example, on an interval $(a, b) \subset (1, \rho)$, one can take

$$\psi_{a,b}(t) = \begin{cases} \exp\left(-\frac{1}{(t-a)(b-t)}\right), & a < t < b, \\ 0, & \text{otherwise.} \end{cases}$$

Choose three linearly independent seeds

$$\varphi_1, \varphi_2, \varphi_3 \in \mathcal{S}.$$

Their images under Λ are three vectors in the two-dimensional space $\mathbb{R}^2$. Hence there exist real coefficients, not all zero, such that

$$c_1\Lambda(\varphi_1) + c_2\Lambda(\varphi_2) + c_3\Lambda(\varphi_3) = 0.$$

Set

$$\varphi = c_1\varphi_1 + c_2\varphi_2 + c_3\varphi_3.$$

Since the seeds are linearly independent, $\varphi \neq 0$, while

$$L_-(\varphi) = L_+(\varphi) = 0. \tag{7.1}$$

Let G_φ be its annular extension, and define

$$g(t) = \begin{cases} G_\varphi(t), & t \neq 0, \\ 0, & t = 0. \end{cases} \tag{7.2}$$

By (7.1), g is continuous at 0. Furthermore, from the equation on the punctured line,

$$G'_\varphi(t) = G_\varphi(\lambda t) - \beta G_\varphi(t) \longrightarrow 0 \quad (t \rightarrow 0), \tag{7.3}$$

because both one-sided limits of G_φ are zero.

For $t \neq 0$, the mean value theorem applied between 0 and t gives some ξ_t between 0 and t such that

$$\frac{g(t) - g(0)}{t} = G'_\varphi(\xi_t).$$

By (7.3), the right-hand side tends to zero. Thus

$$g'(0) = 0.$$

Moreover (g') is continuous at 0, and the differential equation holds there as well:

$$g'(0) = 0 = g(0) - \beta g(0).$$

Therefore

$$g \in F(\beta, \lambda).$$

By Lemma 1, $g \in C^\infty$. Since $g(0) = 0$, Corollary 2 gives

$$g^{(n)}(0) = 0 \quad (n \geq 0).$$

But g is nonzero, because it agrees with the nonzero seed φ on A_0 . A polynomial flat at a point must be the zero polynomial. Hence g is nonpolynomial.

This proves existence for every admissible (β, λ) .

6.6.1 Infinite-dimensionality

The same argument gives more than one solution. Given any $N \geq 1$, choose $N + 2$ linearly independent seeds and restrict Λ to their $(N + 2)$ -dimensional span. Its rank is at most 2, so its kernel has dimension at least N . Since restriction to A_0 recovers the seed, the corresponding global extensions remain linearly independent.

Thus the subspace of flat solutions in $F(\beta, \lambda)$ is infinite-dimensional. $\square$

6.7 8. Analytic solutions

The construction also explains the difference between the C^1 and analytic categories.

Proposition A solution $g \in F(\beta, \lambda)$ that is real analytic in a neighborhood of 0 is necessarily polynomial. More precisely:

- if $\beta \neq \lambda^p$ for every $p \in \mathbb{N}_0$, then $g = 0$;
- if $\beta = \lambda^p$, then g is a scalar multiple of P_p .

Proof From (3.2),

$$g^{(n)}(0) = g(0) \prod_{k=0}^{n-1} (\lambda^k - \beta). \quad (8.1)$$

If $g(0) = 0$, all derivatives at 0 vanish. Analyticity gives $g = 0$ on a neighborhood of 0. If g vanishes on $(-\varepsilon, \varepsilon)$, then

$$g(\lambda t) = g'(t) + \beta g(t) = 0$$

for $|t| < \varepsilon$, so g vanishes on

$$(-|\lambda|\varepsilon, |\lambda|\varepsilon).$$

Iteration gives $g = 0$ on all of $\mathbb{R}$.

Now suppose $g(0) \neq 0$. If no factor in (8.1) vanishes, then for all sufficiently large k ,

$$|\lambda^k - \beta| \geq \frac{1}{2} |\lambda|^k.$$

Consequently the Taylor coefficient

$$\frac{g^{(n)}(0)}{n!}$$

grows in n -th root faster than a constant multiple of

$$\frac{|\lambda|^{n/2}}{n},$$

which tends to $+\infty$. The Taylor series would therefore have radius of convergence zero, contradicting analyticity.

Hence some factor vanishes:

$$\beta = \lambda^p.$$

Then recurrence (8.1) terminates at order p , and the Taylor series is a scalar multiple of P_p . The same outward propagation argument shows equality on all of $\mathbb{R}$. $\square$

Thus all the nonpolynomial solutions constructed above are necessarily nonanalytic at 0; in fact, they are nonzero and flat there.

6.8 9. Boundary and small-case checks

The case $p = 0$ Here $\beta = \lambda^0 = 1$, and

$$P_0(t) = 1.$$

Indeed,

$$P'_0(t) = 0 = P_0(\lambda t) - P_0(t).$$

The theorem nevertheless supplies infinitely many nonpolynomial flat solutions in $F(1, \lambda)$.

The case $p = 1$ Here $\beta = \lambda$, and

$$P_1(t) = 1 + (1 - \lambda)t.$$

Directly,

$$P'_1(t) = 1 - \lambda$$

and

$$P_1(\lambda t) - \lambda P_1(t) = 1 - \lambda.$$

Negative dilations When $\lambda < -1$, the map $t \mapsto \lambda t$ exchanges the two half-lines. This is why the construction imposes the two independent conditions

$$L_-(\varphi) = L_+(\varphi) = 0.$$

No parity assumption and no sign assumption on β are needed.

Why the strict inequality $|\lambda| > 1$ matters The proof uses the geometrically shrinking annuli and the summability

$$\sum_{n \geq 1} (\rho^{-n+1} - \rho^{-n}) = 1.$$

The universal conclusion fails at the boundary.

If $\lambda = 1$, then

$$g' = (1 - \beta)g.$$

For $\beta = 1$, all solutions are constant, so there are no nonpolynomial solutions.

If $\lambda = -1$, differentiating the equation gives

$$g'' = (\beta^2 - 1)g, \quad g'(0) = (1 - \beta)g(0).$$

For $\beta = 1$, all solutions are constant. For $\beta = -1$, all solutions are of the form

$$g(t) = c(1 + 2t),$$

again polynomial. Thus the strict condition $|\lambda| > 1$ cannot simply be replaced by $|\lambda| \geq 1$.

6.8.1 Conclusion

For every real β and every real λ with $|\lambda| > 1$, nonpolynomial solutions not only exist: the space $F(\beta, \lambda)$ contains an infinite-dimensional family of nonzero C^∞ solutions flat at the origin. The polynomial solutions occur exactly at the resonant parameters $\beta = \lambda^p$, but even at those parameters they form only a one-dimensional subspace inside a much larger solution space.

Formal verification. Lean source; online type-check.

7 Question Q759

Informal verdict. A — **Complete solution**, with high confidence.

Lean coverage (partial). Lean proves uniform convergence of every fixed-order derivative for both sequences, which is the full qualitative assertion in the problem. The additional quantitative rates and sharpness claims from the informal answer are not formalized.

7.1 Audited informal answer

7.1.1 Faithful restatement

Let

$$E = C^\infty([-1, 1]; \mathbb{R}), \quad \|h\|_\infty = \max_{x \in [-1, 1]} |h(x)|.$$

For a sequence $(f_n) \subset E$ and $g \in E$, the notation

$$(f_n) \Rightarrow g$$

means

$$\forall k \in \mathbb{N}, \quad \lim_{n \rightarrow \infty} |f_n^{(k)} - g^{(k)}|_\infty = 0.$$

Define

$$\alpha(x) = \begin{cases} \exp\left(-\frac{x^2}{1-x^2}\right), & |x| < 1, \\ 0, & x = \pm 1. \end{cases}$$

Determine whether $f_n \Rightarrow \alpha$ for each of the two sequences

$$f_n^{(a)}(x) = \exp(-x^2 - x^4 - \dots - x^{2n})$$

and

$$f_n^{(b)}(x) = \left(1 - \frac{x^2 + x^4 + \dots + x^{2n}}{n}\right)^n.$$

The visual PDF confirms that in part *b* the whole sum $x^2 + x^4 + \dots + x^{2n}$ is divided by n . There is no apparent mathematical typo. The phrase “equip E with the uniform norm” is to be read together with the displayed definition: $\Rightarrow$ is convergence for all seminorms $h \mapsto |h^{(k)}|_\infty$, i.e. the usual C^∞ -topology.

7.1.2 Final result

Theorem Both sequences converge to α in the sense of the problem:

$$\boxed{f_n^{(a)} \Rightarrow \alpha} \quad \text{and} \quad \boxed{f_n^{(b)} \Rightarrow \alpha}.$$

More precisely, for every $k, M \in \mathbb{N}$,

$$\max_{0 \leq j \leq k} \left\| (f_n^{(a)} - \alpha)^{(j)} \right\|_\infty = O_{k,M}(n^{-M}),$$

whereas, for every fixed k ,

$$\max_{0 \leq j \leq k} \left\| (f_n^{(b)} - \alpha)^{(j)} \right\|_\infty = O_k(n^{-1}).$$

The rate $O(n^{-1})$ in the second assertion is sharp already for uniform convergence of the functions.

7.2 1. Reduction to functions on $([0,1])$

Set

$$P_n(t) = \sum_{j=1}^n t^j, \quad 0 \leq t \leq 1,$$

and, for $0 \leq t < 1$,

$$P(t) = \sum_{j=1}^{\infty} t^j = \frac{t}{1-t}.$$

Define

$$A_n(t) = e^{-P_n(t)}, \quad B_n(t) = \left(1 - \frac{P_n(t)}{n}\right)^n,$$

and

$$\beta(t) = \begin{cases} e^{-P(t)} = e^{-t/(1-t)}, & 0 \leq t < 1, \\ 0, & t = 1. \end{cases}$$

Then

$$f_n^{(a)}(x) = A_n(x^2), \quad f_n^{(b)}(x) = B_n(x^2), \quad \alpha(x) = \beta(x^2).$$

Thus it suffices to prove

$$A_n \longrightarrow \beta \quad \text{and} \quad B_n \longrightarrow \beta \quad \text{in } C^\infty([0, 1]).$$

Indeed, for every smooth U and every $k \geq 0$, repeated differentiation gives

$$\frac{d^k}{dx^k} U(x^2) = k! \sum_{\ell=0}^{\lfloor k/2 \rfloor} \frac{(2x)^{k-2\ell}}{(k-2\ell)!} U^{(k-\ell)}(x^2).$$

Consequently,

$$\left\| \frac{d^k}{dx^k} (U_n(x^2) - U(x^2)) \right\|_{\infty} \leq C_k \max_{0 \leq j \leq k} |U_n^{(j)} - U^{(j)}|_{\infty, [0, 1]}.$$

7.2.1 1.1. Smoothness and flatness of β

For $t < 1$, put

$$u = \frac{1}{1-t}.$$

Then

$$\beta(t) = e^{1-u} = e, e^{-u}.$$

Since $u' = u^2$, induction gives polynomials Q_r such that

$$\beta^{(r)}(t) = \beta(t), Q_r! \left(\frac{1}{1-t} \right), \quad 0 \leq t < 1.$$

Indeed, $Q_0 = 1$, and if the formula holds at order r , then

$$Q_{r+1}(u) = u^2(Q'_r(u) - Q_r(u)).$$

For every polynomial Q ,

$$e^{-u}Q(u) \longrightarrow 0 \quad (u \rightarrow +\infty).$$

Hence

$$\lim_{t \rightarrow 1^-} \beta^{(r)}(t) = 0 \quad \text{for every } r \geq 0.$$

Extending every derivative by the value 0 at $t = 1$ gives a C^∞ function. For example, differentiability at the endpoint follows successively from the mean value theorem. Thus

$$\beta \in C^\infty([0, 1]), \quad \beta^{(r)}(1) = 0 \quad (r \geq 0).$$

In particular, $\alpha = \beta \circ (x \mapsto x^2)$ belongs to E and is flat at both endpoints.

7.3 2. The first sequence

We prove

$$A_n = e^{-P_n} \longrightarrow \beta = e^{-P} \quad \text{in } C^\infty([0, 1]).$$

7.3.1 2.1. Elementary derivative bounds

For $r \geq 0$ and $0 \leq t \leq 1$,

$$|P_n^{(r)}(t)| \leq n^{r+1}.$$

For $r = 0$ this is $P_n(t) \leq n$. For $r \geq 1$,

$$P_n^{(r)}(t) = \sum_{j=r}^n (j)_r t^{j-r},$$

where

$$(j)_r = j(j-1) \cdots (j-r+1),$$

and hence

$$|P_n^{(r)}(t)| \leq \sum_{j=1}^n j^r \leq n^{r+1}.$$

Repeated use of the product and chain rules shows that every term in $A_n^{(r)}$ is of the form

$$c, e^{-P_n} \prod_{i=1}^m P_n^{(s_i)},$$

where

$$s_i \geq 1, \quad s_1 + \cdots + s_m = r, \quad m \leq r.$$

Therefore

$$\boxed{|A_n^{(r)}(t)| \leq C_r n^{2r} e^{-P_n(t)} \leq C_r n^{2r}.} \quad (2.1)$$

7.3.2 2.2. Boundary layer

Let

$$\rho_n = n^{-1/2}, \quad J_n = [1 - \rho_n, 1].$$

We first show that

$$P_n(t) \geq c\sqrt{n} \quad (t \in J_n) \quad (2.2)$$

for an absolute $c > 0$ and all sufficiently large n .

Choose

$$m = \left\lfloor \frac{1}{4\rho_n} \right\rfloor.$$

Then $m \leq n$ for large n . If $t \geq 1 - \rho_n$ and $1 \leq j \leq m$, then, using

$$\log(1 - s) \geq -2s \quad (0 \leq s \leq \tfrac{1}{2}),$$

we get

$$t^j \geq (1 - \rho_n)^m \geq e^{-2m\rho_n} \geq e^{-1/2}.$$

Moreover $m \geq (8\rho_n)^{-1}$ for large n . Thus

$$P_n(t) \geq \sum_{j=1}^m t^j \geq \frac{e^{-1/2}}{8\rho_n} = c\sqrt{n}.$$

Combining (2.1) and (2.2),

$$\sup_{t \in J_n} |A_n^{(r)}(t)| \leq C_r n^{2r} e^{-c\sqrt{n}}. \quad (2.3)$$

For β , the representation

$$\beta^{(r)}(t) = \beta(t) Q_r! \left(\frac{1}{1-t} \right)$$

and $1/(1-t) \geq \sqrt{n}$ on J_n imply

$$\sup_{t \in J_n} |\beta^{(r)}(t)| \leq C_r e^{-c_r \sqrt{n}}. \quad (2.4)$$

Thus all derivatives of both A_n and β are uniformly negligible in the boundary layer.

7.3.3 2.3. Interior region

Let

$$I_n = [0, 1 - \rho_n].$$

For $t < 1$,

$$P(t) - P_n(t) = \sum_{j=n+1}^{\infty} t^j = \frac{t^{n+1}}{1-t} =: R_n(t).$$

Hence

$$\beta(t) = A_n(t) e^{-R_n(t)}$$

and

$$A_n(t) - \beta(t) = A_n(t) (1 - e^{-R_n(t)}). \quad (2.5)$$

For $r \geq 0$, Leibniz's rule applied to

$$R_n(t) = t^{n+1} (1-t)^{-1}$$

gives

$$R_n^{(r)}(t) = \sum_{a=0}^r \binom{r}{a} (n+1)_a t^{n+1-a} (r-a)! (1-t)^{-(r-a+1)}.$$

On I_n , $1-t \geq \rho_n$ and $t \leq 1 - \rho_n$. Therefore

$$|R_n^{(r)}(t)| \leq C_r n^r \rho_n^{-(r+1)} (1 - \rho_n)^{n+1-r}.$$

Since

$$(1 - \rho_n)^{n+1-r} \leq e^{-\rho_n(n+1-r)} \leq e^{-\sqrt{n}/2}$$

for large n , we obtain

$$\max_{0 \leq r \leq k} |R_n^{(r)}|_{\infty, I_n} \leq C_k n^{C_k} e^{-\sqrt{n}/2}. \quad (2.6)$$

Every derivative of $1 - e^{-R_n}$ is a finite sum of products of derivatives of R_n , with at least one such factor. Thus (2.6) implies

$$\max_{0 \leq r \leq k} \left| (1 - e^{-R_n})^{(r)} \right|_{\infty, I_n} \leq C_k n^{C_k} e^{-c\sqrt{n}}. \quad (2.7)$$

Applying Leibniz's rule to (2.5), using (2.1) and (2.7), gives

$$\max_{0 \leq r \leq k} |A_n^{(r)} - \beta^{(r)}|_{\infty, I_n} \leq C_k n^{C_k} e^{-c\sqrt{n}}. \quad (2.8)$$

Together with the boundary estimates (2.3)–(2.4), this proves

$$\boxed{|A_n - \beta|_{C^k([0,1])} \leq C_k n^{C_k} e^{-c\sqrt{n}}.} \quad (2.9)$$

In particular, for every M ,

$$|A_n - \beta|_{C^k([0,1])} = O_{k,M}(n^{-M}).$$

After composition with x^2 , this proves part *a*:

$$\boxed{f_n^{(a)} \Rightarrow \alpha.}$$

7.4 3. A weighted binomial approximation

To handle the second sequence uniformly up to the endpoint, we need a version of

$$\left(1 - \frac{y}{n}\right)^n \longrightarrow e^{-y}$$

that remains valid after differentiation and permits y to range over $([0,n])$.

7.4.1 Lemma 3.1

For every pair of nonnegative integers (m,L) , there is a constant $C_{m,L}$ such that, for all sufficiently large n ,

$$\boxed{\sup_{0 \leq y \leq n} (1+y)^L \left| \frac{d^m}{dy^m} \left(1 - \frac{y}{n}\right)^n - \frac{d^m}{dy^m} e^{-y} \right| \leq \frac{C_{m,L}}{n}.} \quad (3.1)$$

Proof For $n \geq m$,

$$\frac{d^m}{dy^m} \left(1 - \frac{y}{n}\right)^n = (-1)^m c_{n,m} \left(1 - \frac{y}{n}\right)^{n-m},$$

where

$$c_{n,m} = \frac{(n)_m}{n^m} = \prod_{j=0}^{m-1} \left(1 - \frac{j}{n}\right).$$

Also

$$\frac{d^m}{dy^m} e^{-y} = (-1)^m e^{-y}.$$

Since all factors in $c_{n,m}$ lie in $([0,1])$,

$$0 \leq 1 - c_{n,m} \leq \sum_{j=0}^{m-1} \frac{j}{n} \leq \frac{C_m}{n}. \quad (3.2)$$

Put

$$h_{n,m}(y) = \left(1 - \frac{y}{n}\right)^{n-m}.$$

Region $0 \leq y \leq n/2$ Set $z = y/n$ and

$$a = -(n-m) \log(1-z),$$

so that $h_{n,m}(y) = e^{-a}$. For $n \geq 2m$,

$$a \geq (n-m)z \geq \frac{y}{2}.$$

Moreover,

$$a - y = (n-m)(-\log(1-z) - z) - mz.$$

For $0 \leq z \leq 1/2$,

$$0 \leq -\log(1-z) - z = \sum_{\ell=2}^{\infty} \frac{z^\ell}{\ell} \leq \frac{z^2}{1-z} \leq 2z^2.$$

Consequently,

$$|a - y| \leq C_m \left(\frac{y}{n} + \frac{y^2}{n} \right).$$

By the mean value theorem applied to $s \mapsto e^{-s}$,

$$|e^{-a} - e^{-y}| \leq e^{-\min(a,y)} |a - y| \leq \frac{C_m}{n} (y + y^2) e^{-y/2}.$$

Multiplication by $(1+y)^L$ preserves the $O(1/n)$ bound, because every polynomial times $e^{-y/2}$ is bounded on $[0, \infty)$.

Also

$$h_{n,m}(y) \leq e^{-y/2},$$

so (3.2) gives the same weighted $O(1/n)$ estimate for

$$(c_{n,m} - 1)h_{n,m}(y).$$

Region $n/2 \leq y \leq n$ Here

$$h_{n,m}(y) \leq 2^{-(n-m)}, \quad e^{-y} \leq e^{-n/2}.$$

Thus

$$(1+y)^L (h_{n,m}(y) + e^{-y}) \leq (1+n)^L (2^{-(n-m)} + e^{-n/2}) = O_{m,L}(n^{-1}).$$

Combining both regions and (3.2) proves the lemma. $\square$

7.5 4. Derivatives of the geometric partial sums

The following estimate allows us to compose Lemma 3.1 with P_n .

7.5.1 Lemma 4.1

For every $r \geq 0$, there is $C_r > 0$, independent of n , such that

$$\boxed{|P_n^{(r)}(t)| \leq C_r (1 + P_n(t))^{r+1}} \tag{4.1}$$

for all $n \geq 1$ and $0 \leq t \leq 1$.

Proof The case $r = 0$ is immediate. Suppose $r \geq 1$.

If $0 \leq t \leq 1/2$, then

$$P_n^{(r)}(t) \leq \sum_{j=r}^{\infty} (j)_r 2^{-(j-r)} =: C_r,$$

and (4.1) follows.

Now suppose $t > 1/2$, and set

$$\delta = 1 - t, \quad L = \min(n, \delta^{-1}).$$

First,

$$P_n^{(r)}(t) \leq \sum_{j=1}^n j^r \leq C_r n^{r+1}. \quad (4.2)$$

Also, since $t^{-r} \leq 2^r$ and $t^j \leq e^{-\delta j}$,

$$P_n^{(r)}(t) \leq 2^r \sum_{j=1}^{\infty} j^r e^{-\delta j} \leq C_r \delta^{-(r+1)}. \quad (4.3)$$

Therefore

$$P_n^{(r)}(t) \leq C_r L^{r+1}. \quad (4.4)$$

It remains to compare L to $P_n(t)$. If $L \leq 8$, then trivially

$$L \leq 8(1 + P_n(t)).$$

If $L > 8$, choose

$$m = \left\lfloor \frac{L}{4} \right\rfloor.$$

Then

$$m \geq \frac{L}{8}, \quad m \leq n, \quad m\delta \leq \frac{1}{4}.$$

For $1 \leq j \leq m$,

$$t^j = (1 - \delta)^j \geq e^{-2\delta j} \geq e^{-1/2}.$$

Hence

$$P_n(t) \geq \sum_{j=1}^m t^j \geq e^{-1/2} m \geq cL.$$

Combining this with (4.4) proves (4.1). $\square$

7.6 5. The second sequence

Let

$$G_n(y) = \left(1 - \frac{y}{n}\right)^n, \quad 0 \leq y \leq n,$$

and

$$G(y) = e^{-y}.$$

Then

$$B_n = G_n \circ P_n, \quad A_n = G \circ P_n.$$

We prove that for every fixed k ,

$$|B_n - A_n|_{C^k([0,1])} = O_k(n^{-1}). \quad (5.1)$$

For $k = 0$, Lemma 3.1 with $m = L = 0$ immediately gives

$$|B_n - A_n|_\infty \leq \frac{C}{n}.$$

Now let $k \geq 1$. Repeated application of the chain rule shows that the k -th derivative of

$$(G_n - G) \circ P_n$$

is a finite linear combination, with coefficients depending only on k , of terms

$$(G_n^{(m)} - G^{(m)})(P_n(t)) \prod_{i=1}^m P_n^{(r_i)}(t),$$

where

$$1 \leq m \leq k, \quad r_i \geq 1, \quad r_1 + \cdots + r_m = k.$$

By Lemma 4.1,

$$\prod_{i=1}^m |P_n^{(r_i)}(t)| \leq C_k (1 + P_n(t))^{\sum_i (r_i + 1)} = C_k (1 + P_n(t))^{k+m}.$$

Because $m \leq k$,

$$\prod_{i=1}^m |P_n^{(r_i)}(t)| \leq C_k (1 + P_n(t))^{2k}. \quad (5.2)$$

Applying Lemma 3.1 with weight exponent $L = 2k$ gives

$$\left| G_n^{(m)}(P_n(t)) - G^{(m)}(P_n(t)) \right| \leq \frac{C_k}{n(1 + P_n(t))^{2k}}. \quad (5.3)$$

The product of (5.2) and (5.3) is $O_k(1/n)$, uniformly in t . There are only finitely many chain-rule terms, so (5.1) follows.

From the first part,

$$|A_n - \beta|_{C^k([0,1])} = O_{k,M}(n^{-M})$$

for every M . Taking $M = 2$, for example,

$$\boxed{|B_n - \beta|_{C^k([0,1])} \leq \frac{C_k}{n}}. \quad (5.4)$$

Composing with x^2 now yields

$$\boxed{f_n^{(b)} \Rightarrow \alpha}.$$

7.7 6. Boundary checks, small cases, and sharpness

7.7.1 6.1. At $x = 0$

For every n ,

$$f_n^{(a)}(0) = f_n^{(b)}(0) = \alpha(0) = 1.$$

All three functions are even, so all odd derivatives at 0 vanish.

For the first sequence, the agreement is much stronger. Near 0,

$$\sum_{j=n+1}^{\infty} x^{2j} = O(x^{2n+2}),$$

and therefore

$$f_n^{(a)}(x) - \alpha(x) = O(x^{2n+2}).$$

Thus the derivatives at 0 agree exactly through order $2n + 1$.

7.7.2 6.2. At $x = \pm 1$

Since α is flat,

$$\alpha^{(k)}(\pm 1) = 0 \quad (k \geq 0).$$

For the first sequence,

$$f_n^{(a)}(\pm 1) = e^{-n}.$$

Every fixed-order derivative at ± 1 is e^{-n} times a polynomial in n , and hence tends to 0.

For the second sequence,

$$1 - \frac{x^2 + x^4 + \cdots + x^{2n}}{n}$$

has a simple zero at each of $x = \pm 1$. Hence $f_n^{(b)}$ has a zero of order n at each endpoint. In particular,

$$(f_n^{(b)})^{(k)}(\pm 1) = 0 \quad \text{whenever } n > k.$$

7.7.3 6.3. Sharpness for part b

Fix $0 < t < 1$. Since $P_n(t) \rightarrow P(t)$ exponentially fast and remains bounded,

$$n \log \left(1 - \frac{P_n(t)}{n} \right) = -P_n(t) - \frac{P_n(t)^2}{2n} + O_t(n^{-2}).$$

It follows that

$$B_n(t) = e^{-P_n(t)} \left(1 - \frac{P_n(t)^2}{2n} + O_t(n^{-2}) \right).$$

Consequently,

$$\boxed{n(B_n(t) - \beta(t)) \longrightarrow -\frac{1}{2}P(t)^2 e^{-P(t)}}. \quad (6.1)$$

For example, take $t = 1/2$, corresponding to $x = 1/\sqrt{2}$. Then

$$P(1/2) = 1, \quad \beta(1/2) = e^{-1},$$

and

$$n \left(f_n^{(b)}! \left(\frac{1}{\sqrt{2}} \right) - \alpha \left(\frac{1}{\sqrt{2}} \right) \right) \longrightarrow -\frac{1}{2e}.$$

Therefore

$$|f_n^{(b)} - \alpha|_\infty \neq o(n^{-1}),$$

so the $O(n^{-1})$ rate is optimal.

7.7.4 6.4. Role of the domain

The restriction to $([-1,1])$ is essential. If $|x| > 1$, then

$$x^2 + x^4 + \cdots + x^{2n} \asymp_x x^{2n},$$

so

$$\left| 1 - \frac{x^2 + x^4 + \cdots + x^{2n}}{n} \right|^n \longrightarrow +\infty.$$

Thus the second sequence has no analogous convergence on any interval containing a point with $|x| > 1$.

The endpoint values $\alpha(\pm 1) = 0$ are forced by the interior limit and, more strongly, make α flat and C^∞ at the endpoints.

7.8 Conclusion

For both proposed approximation schemes, not only the functions but every fixed derivative converges uniformly on the full closed interval:

$$\boxed{\forall k \in \mathbb{N}, \quad \lim_{n \rightarrow \infty} \left\| (f_n^{(a)})^{(k)} - \alpha^{(k)} \right\|_{\infty} = 0,}$$

and

$$\boxed{\forall k \in \mathbb{N}, \quad \lim_{n \rightarrow \infty} \left\| (f_n^{(b)})^{(k)} - \alpha^{(k)} \right\|_{\infty} = 0.}$$

Hence the answer to both parts is affirmative.

Formal verification. Lean source; online type-check.

8 Question Q764

Informal verdict. A — **Complete solution**, with high confidence.

Lean coverage (partial). Lean proves the line block-cost lemmas and dynamic-program feasibility recurrence, existence of an optimum in finite metrics, the farthest-first factor-two theorem, and the 1–2 metric lemma. Computational-complexity and hardness claims are not formalized.

8.1 Audited informal answer

8.1.1 1. Faithful restatement

Let $1 \leq k \leq n$.

a Given distinct real numbers

$$X = x_1 < x_2 < \cdots < x_n,$$

and $F \subseteq X$, define

$$\delta(X, F) = \max_{x \in X} \min_{y \in F} |x - y|.$$

Find an efficient algorithm producing a k -element subset $F \subseteq X$ minimizing $\delta(X, F)$.

b Solve the analogous problem when X is an arbitrary n -point subset of a metric space, with $(|x-y|)$ replaced by the metric distance.

There are three minor ambiguities in the wording.

1. The statement only says $k \leq n$, but k must be a positive integer, since the minimum over an empty F is otherwise undefined.
2. The minimizing subset need not be unique, so “the subset F ” should be read as “a minimizing subset F .”
3. “Efficient” is not formally defined. Below, arithmetic and comparisons of input distances are counted at unit cost. For rational inputs, the algorithms also have polynomial bit complexity.

8.1.2 2. Final result

Theorem

1. **Ordered real line.** An optimal k -element set can be found exactly in

$$O(n^2 + kn) = O(n^2)$$

time and $O(n^2 + kn) = O(n^2)$ memory. The algorithm is a bottleneck dynamic program over consecutive blocks.

1. **Arbitrary finite metric.**

- The problem is exactly the finite **discrete metric k -center problem**.
- Its decision version is NP-complete, even when every nonzero distance is either 1 or 2. Consequently, unless $P = NP$, there is no polynomial-time exact algorithm when k is part of the input.
- An exact exhaustive algorithm takes

$$O\left(\binom{n}{k} nk\right)$$

time.

- There is an $O(nk)$ -time polynomial algorithm returning a set F with

$$\delta(X, F) \leq 2, \delta_{\text{opt}}.$$

Unless $P = NP$, no polynomial algorithm can guarantee a factor strictly smaller than 2.

8.2 Part a: points on the real line

8.2.1 3. The optimal cost of one consecutive block

For $1 \leq i \leq j \leq n$, define

$$c(i, j) = \min_{i \leq \ell \leq j} \max_{i \leq t \leq j} |x_t - x_\ell|.$$

Thus $c(i, j)$ is the smallest radius with which one center, constrained to be one of

$$x_i, x_{i+1}, \dots, x_j,$$

can cover that whole block.

For a fixed center x_ℓ , the farthest point of the block is one of its two endpoints, so

$$\max_{i \leq t \leq j} |x_t - x_\ell| = \max \{x_\ell - x_i, x_j - x_\ell\}.$$

Writing

$$m_{ij} = \frac{x_i + x_j}{2},$$

we have

$$\max \{x_\ell - x_i, x_j - x_\ell\} = \frac{x_j - x_i}{2} + |x_\ell - m_{ij}|.$$

Consequently:

Lemma 1 An optimal one-center location for the block $x_i, \dots, x_j$ is any x_ℓ nearest to the midpoint $(x_i + x_j)/2$, and

$$c(i, j) = \frac{x_j - x_i}{2} + \min_{i \leq \ell \leq j} \left| x_\ell - \frac{x_i + x_j}{2} \right|. \quad (1)$$

In particular, the center is constrained to lie in X ; this is not the continuous k -center problem.

8.2.2 4. Reduction to consecutive blocks

The order on the real line provides the crucial structure.

Lemma 2: consecutive-cell lemma For every set of centers

$$F = f_1 < \dots < f_s \subseteq X,$$

the points of X can be assigned to nearest centers so that the points assigned to each f_t form a consecutive block of X .

Proof Break ties consistently, for example in favor of the smaller center. The points at least as close to f_t as to f_{t+1} lie to the left of the midpoint

$$\frac{f_t + f_{t+1}}{2}.$$

Thus the nearest-center index is nondecreasing as one moves from left to right through X . Its level sets are therefore consecutive intervals of indices.

Every center belongs to its own cell, since its distance to itself is zero, so all cells are nonempty. $\square$

It follows that an optimal solution is equivalent to partitioning X into at most k consecutive nonempty blocks and placing an optimal one-center solution in every block.

More precisely, if

$$0 = i_0 < i_1 < \cdots < i_s = n,$$

the associated blocks are

$$x_{i_{t-1}+1}, \dots, x_{i_t}, \quad 1 \leq t \leq s.$$

Proposition 3: bottleneck-partition characterization The optimal value is

$$R_k = \min_{\substack{1 \leq s \leq k \\ 0=i_0 < i_1 < \cdots < i_s=n}} \max_{1 \leq t \leq s} c(i_{t-1} + 1, i_t). \quad (2)$$

Proof Given any set of at most k centers, Lemma 2 produces at most k consecutive cells. On a cell $([i, j])$, its chosen center has radius at least $c(i, j)$. Therefore its total radius is at least the right-hand side of 2.

Conversely, given a partition occurring on the right-hand side, choose in every block a center realizing $c(i, j)$. Every point is then within the displayed maximum radius of its block center. $\square$

Using fewer than k centers causes no difficulty: arbitrary unused points of X can be added as extra centers, and doing so cannot increase the radius. Thus the minimum over “at most k ” and the minimum over “exactly k ” coincide.

8.2.3 5. Dynamic programming recurrence

Let

$$\rho_p(j)$$

be the optimal radius for the prefix

$$x_1, \dots, x_j$$

using at most p centers. Set

$$\rho_p(0) = 0, \quad \rho_0(j) = +\infty \quad (j \geq 1).$$

The last block of a partition of $x_1, \dots, x_j$ must have the form

$$x_i, x_{i+1}, \dots, x_j$$

for some $1 \leq i \leq j$. Hence:

Lemma 4: dynamic-programming recurrence For $p, j \geq 1$,

$$\boxed{\rho_p(j) = \min_{1 \leq i \leq j} \max\{\rho_{p-1}(i-1), c(i, j)\}.} \quad (3)$$

Proof If the last block starts at i , the preceding prefix uses at most $p-1$ centers and has optimal cost $\rho_{p-1}(i-1)$; the final block costs $c(i, j)$. The total bottleneck cost is their maximum.

Conversely, every partition counted by $\rho_p(j)$ has a uniquely determined last-block starting index i , so every possible solution appears in the minimum. $\square$

A direct evaluation of 3 would take $O(kn^2)$ time. We now exploit monotonicity to reduce the dynamic-programming part to $O(kn)$.

8.2.4 6. Monotonicity of the transition

For fixed p and j , put

$$A_i = \rho_{p-1}(i-1), \quad B_i = c(i, j).$$

Lemma 5 As i increases:

1. A_i is nondecreasing;
2. B_i is nonincreasing.

Moreover, for fixed i , $c(i, j)$ is nondecreasing in j .

Proof The value A_i concerns a prefix ending at $i - 1$. Enlarging that prefix cannot decrease the required radius, so A_i is nondecreasing.

For B_i , removing the leftmost point of a block cannot increase the optimum. If an optimal center for $([i, j])$ is not x_i , retain it. If it is x_i , replace it by x_{i+1} ; every remaining point lies to the right, and is no farther from x_{i+1} than from x_i . Thus

$$c(i + 1, j) \leq c(i, j).$$

Similarly, adding a point on the right cannot reduce the one-center cost. If an optimal center for $([i, j+1])$ is one of $x_i, \dots, x_j$, restricting to the smaller block proves the assertion. If it is x_{j+1} , then x_j is at least as good for all points $x_i, \dots, x_j$. Hence

$$c(i, j) \leq c(i, j + 1).$$

□

Let

$$t = t_p(j) = \min i \in 1, \dots, j : A_i \geq B_i,$$

with the convention $t = j + 1$ if this set is empty.

For $i < t$,

$$\max(A_i, B_i) = B_i,$$

which is nonincreasing in i . For $i \geq t$,

$$\max(A_i, B_i) = A_i,$$

which is nondecreasing in i . Therefore:

Lemma 6: crossing lemma A minimizing index in 3 is always one of

$$t_p(j) - 1, \quad t_p(j),$$

after discarding indices outside $1, \dots, j$.

Furthermore, for fixed p ,

$$t_p(1) \leq t_p(2) \leq \dots \leq t_p(n). \tag{4}$$

Proof The first assertion follows from the decreasing-then-increasing behavior just described.

For the second, if $i < t_p(j)$, then

$$A_i < c(i, j) \leq c(i, j + 1),$$

so $i < t_p(j + 1)$. Hence the first crossing cannot move to the left. $\square$

Thus, while j runs from 1 to n , a single pointer tracking $t_p(j)$ moves right at most n times.

8.2.5 7. Precomputing all block costs in $O(n^2)$

By 1, the optimal center for $([i, j])$ is a point nearest to

$$m_{ij} = \frac{x_i + x_j}{2}.$$

For a fixed i , these midpoints are nondecreasing as j increases. If ties are resolved in favor of the rightmost nearest point, the corresponding optimal-center index is also nondecreasing.

Indeed, between consecutive points $x_\ell, x_{\ell+1}$, the point $x_{\ell+1}$ is at least as close as x_ℓ to a number m exactly when

$$m \geq \frac{x_\ell + x_{\ell+1}}{2}.$$

Once this becomes true, it remains true as m moves to the right.

Therefore a moving pointer computes all $c(i, j)$, for fixed i , in $O(n)$ time. Repeating for all i gives $O(n^2)$.

8.2.6 8. Exact quadratic-time algorithm

The following pseudocode uses one-based indices. $C[i, j]$ stores $c(i, j)$, and $M[i, j]$ stores a corresponding optimal center index.

QUADRATIC-LINE-K-CENTER($x[1..n]$, k)

Phase 1: all one-block costs and centers

for $i = 1, \dots, n$:

$ell = i$

 for $j = i, \dots, n$:

 # Move ell to the rightmost point nearest to $(x[i] + x[j])/2$.

```

# The doubled comparison avoids division by 2.
while ell < j and
    abs(2*x[ell+1] - x[i] - x[j])
    <= abs(2*x[ell] - x[i] - x[j]):
    ell = ell + 1

M[i,j] = ell
C[i,j] = max(x[ell] - x[i], x[j] - x[ell])

# Phase 2: bottleneck dynamic programming
prev[0] = 0
for j = 1,...,n:
    prev[j] = +infinity

for p = 1,...,k:
    cur[0] = 0
    t = 1

    for j = 1,...,n:
        # After this loop, t is the first crossing index;
        # if no crossing exists, t = j+1.
        while t <= j and prev[t-1] < C[t,j]:
            t = t + 1

        candidates = empty set
        if t <= j:
            insert t into candidates
        if t >= 2:
            insert t-1 into candidates

        choose i in candidates minimizing
            max(prev[i-1], C[i,j])

        cur[j] = max(prev[i-1], C[i,j])
        parent[p,j] = i

    prev = cur

```

```

# Phase 3: reconstruct the consecutive blocks
F = empty set
p = k
j = n

while j > 0:
    i = parent[p,j]
    insert x[M[i,j]] into F
    j = i - 1
    p = p - 1

# The dynamic program may have used fewer than k nonempty blocks.
while |F| < k:
    insert any point of X \ F into F

return F and prev[n]

```

Theorem 7: correctness and complexity The algorithm returns a k -element set $F \subseteq X$ satisfying

$$\delta(X, F) = R_k.$$

Its running time is

$$O(n^2 + kn),$$

and its memory requirement is

$$O(n^2 + kn).$$

Proof The first phase correctly computes every $c(i, j)$ and a realizing center by Lemma 1 and the monotone-midpoint argument.

For each fixed p , the pointer $\mathbf{t}$ is exactly the crossing index from Lemma 6. Since it is nondecreasing in j , it makes at most n increments. Lemma 6 shows that checking $\mathbf{t}-1$ and $\mathbf{t}$ evaluates recurrence 3 exactly. Thus the dynamic program computes every $\rho_p(j)$, in particular

$$\text{prev}[\mathbf{n}] = \rho_k(n) = R_k.$$

Following the parent pointers reconstructs a partition realizing this value. The center $\mathbf{M}[\mathbf{i}, \mathbf{j}]$ realizes the one-block cost for each reconstructed block. Hence the resulting set of centers covers X with radius R_k .

If fewer than k centers occur, adding arbitrary unused points cannot increase the radius. It cannot decrease it below R_k , since R_k was already the minimum over all sets of at most k centers. Thus the final k -element set remains optimal.

The cost table takes $O(n^2)$ time and space. For every p , the dynamic-programming scan takes $O(n)$, giving $O(kn)$. The parent table uses $O(kn)$ space. $\square$

8.2.7 9. A simpler independent decision algorithm

For a proposed radius r , one can decide in linear time whether k centers suffice.

Take the leftmost uncovered point x_i . Choose as center the rightmost x_j satisfying

$$x_j \leq x_i + r.$$

This center covers all points through $x_j + r$. Repeat with the next uncovered point.

This greedy rule uses the minimum possible number of radius- r centers. Indeed, every feasible cover must have some center $x_\ell \leq x_i + r$ covering x_i . The greedy center satisfies $x_j \geq x_\ell$, so

$$x_\ell + r \leq x_j + r;$$

the competing center cannot cover any point farther to the right than the greedy center. Induction on the uncovered suffix proves optimality.

Since an optimal radius is necessarily one of the pairwise distances

$$x_j - x_i,$$

sorting all $O(n^2)$ candidate distances and binary-searching with this decision procedure gives a simpler $O(n^2 \log n)$ -time exact algorithm. The preceding dynamic program removes the logarithmic factor.

8.3 Part b: an arbitrary finite metric space

Let $((X, d))$ be a finite metric space with $|X| = n$, and write

$$d(x, F) = \min_{f \in F} d(x, f).$$

8.3.1 10. Threshold-graph characterization

For $r \geq 0$, define a graph G_r on vertex set X by declaring distinct (u,v) adjacent when

$$d(u, v) \leq r.$$

A subset $F \subseteq X$ satisfies

$$\delta(X, F) \leq r$$

if and only if every vertex is either in F or adjacent in G_r to a vertex of F . In other words, F is a dominating set of G_r .

Let $\gamma(G)$ denote the minimum size of a dominating set of G , and let

$$D = \{d(x, y) : x, y \in X\}.$$

Because X is finite, $\delta(X, F)$ is always an element of D . Therefore:

Proposition 8 The exact optimal radius is

$$\boxed{R_k^{\text{met}} = \min r \in D : \gamma(G_r) \leq k.} \quad (5)$$

Moreover, the optimal k -element sets are precisely the k -element dominating sets of $G_{R_k^{\text{met}}}$.

This is a complete mathematical characterization, but finding a minimum dominating set is computationally difficult.

8.3.2 11. An exact finite algorithm

Assuming the distance matrix is available, one may simply enumerate all k -element subsets:

```

EXACT-METRIC-K-CENTER(X, d, k)

best_radius = +infinity
best_set = none

for every k-element subset F of X:
    r = 0
    for every x in X:
        r = max(r, min_{f in F} d(x, f))

    if r < best_radius:

```

```

    best_radius = r
    best_set = F

return best_set and best_radius

```

There are $\binom{n}{k}$ subsets, and evaluating one requires $O(nk)$ distance comparisons. Thus the running time is

$$O\left(\binom{n}{k}nk\right). \quad (6)$$

For every fixed k , this is polynomial in n , but it is exponential when k is part of the input.

8.3.3 12. NP-completeness of the general metric problem

Consider the decision problem:

Given a finite metric $((X,d))$, an integer k , and a radius r , does there exist $F \subseteq X$, $|F| \leq k$, such that $\delta(X, F) \leq r$?

It belongs to NP: a proposed F can be checked in $O(nk)$ time.

We prove NP-hardness using the standard NP-completeness of **VERTEX COVER**, one of Karp's classical NP-complete problems. ([Informatique Cornell Bowers][1])

Reduction Let $G = (V, E)$ be a nontrivial instance of VERTEX COVER, with integer k . Construct a graph H whose vertex set is the disjoint union

$$W = V \sqcup E,$$

where each edge $e \in E$ is regarded as a new “edge-vertex.”

The adjacencies of H are:

1. all vertices of V are pairwise adjacent, so V induces a clique;
2. an edge-vertex $e = u, v$ is adjacent exactly to u and v ;
3. distinct edge-vertices are not adjacent.

Now define a distance on W by

$$d_H(a, b) = \begin{cases} 0, & a = b, \\ 1, & a \neq b \text{ and } a, b \text{ are adjacent in } H, \\ 2, & \text{otherwise.} \end{cases} \quad (7)$$

Lemma 9 The function d_H is a metric.

Proof Symmetry and positive definiteness are immediate. For the triangle inequality, all nonzero distances are at least 1, while every distance is at most 2. Thus, for three pairwise distinct points,

$$d_H(a, c) \leq 2 \leq d_H(a, b) + d_H(b, c).$$

Cases with repeated points are immediate. $\square$

A set has covering radius at most 1 in this metric exactly when it is a dominating set of H .

Lemma 10 The original graph G has a vertex cover of size at most k if and only if H has a dominating set of size at most k .

Proof Suppose $C \subseteq V$ is a vertex cover. Every edge-vertex $e = u, v$ is adjacent to at least one point of C . Since V is a clique, every vertex of V is in C or adjacent to every member of C . Thus C dominates H .

Conversely, let D dominate H . Whenever an edge-vertex

$$e = u, v$$

belongs to D , replace it by either endpoint, say u . Repeating this operation produces a set

$$C \subseteq V, \quad |C| \leq |D|.$$

Every edge-vertex e is incident to a vertex of C :

- if $e \in D$, an endpoint was inserted when e was replaced;
- if $e \notin D$, then e had to be dominated by one of its endpoints in D , since edge-vertices are not adjacent to one another.

Therefore C contains an endpoint of every edge of G , so C is a vertex cover. $\square$

The construction is polynomial. Thus the metric k -center decision problem is NP-hard even for the fixed threshold $r = 1$ and metrics whose nonzero distances belong to $(\{1, 2\})$. Together with membership in NP:

Theorem 11 The decision version of the arbitrary-metric problem is NP-complete, even for 1-2 metrics.

Consequently, the exact optimization problem in Q764b is NP-hard. Unless $P = NP$, there is no polynomial-time exact algorithm for arbitrary finite metrics when k is part of the input.

The distinction between “at most k ” and “exactly k ” is immaterial: a dominating set of size smaller than k can be augmented by arbitrary vertices.

8.3.4 13. A polynomial-time factor-2 algorithm

Although exact optimization is NP-hard, there is a very simple optimal-threshold approximation.

Farthest-first algorithm Choose an arbitrary first center c_1 . Having selected

$$C_t = c_1, \dots, c_t,$$

choose

$$c_{t+1} \in \arg \max_{x \in X} d(x, C_t).$$

FARTHEST-FIRST(X, d, k)

choose any $c[1]$ in X

$F = \{c[1]\}$

for every x in X :

$\text{nearest}[x] = d(x, c[1])$

for $t = 2, \dots, k$:

 choose $c[t]$ maximizing $\text{nearest}[x]$

 insert $c[t]$ into F

 for every x in X :

$\text{nearest}[x] = \min(\text{nearest}[x], d(x, c[t]))$

return F

It performs $O(n)$ work for each new center, hence $O(nk)$ distance operations.

Theorem 12 If F is returned by farthest-first and R^* is the exact optimal radius, then

$$\delta(X, F) \leq 2R^*.$$

Proof Let

$$r = \delta(X, F),$$

and choose $c_{k+1} \in X$ satisfying

$$d(c_{k+1}, F) = r.$$

We claim that the $k + 1$ points

$$c_1, c_2, \dots, c_k, c_{k+1}$$

are pairwise at distance at least r .

Indeed, when c_t was selected, its distance from the previously selected centers was the then-current maximum covering distance. These maximum distances form a nonincreasing sequence, whose final value is r . Hence for $s < t \leq k$,

$$d(c_s, c_t) \geq r.$$

The same assertion holds for c_{k+1} by its definition.

Now let F^* be an optimal set of k centers with radius R^* . Assign each of the $k + 1$ displayed points to an optimal center at distance at most R^* . By the pigeonhole principle, two of them, say c_s, c_t , are assigned to the same optimal center f . The triangle inequality gives

$$d(c_s, c_t) \leq d(c_s, f) + d(f, c_t) \leq 2R^*.$$

Since $d(c_s, c_t) \geq r$, it follows that

$$r \leq 2R^*.$$

□

8.3.5 14. The factor 2 cannot be improved in polynomial time

The same reduction used for NP-hardness gives a gap instance.

For the 1-2 metric constructed from a VERTEX COVER instance:

- if a vertex cover of size at most k exists, then the optimal metric k -center radius is 1;

- otherwise, no radius-1 solution exists, while every nonempty set of centers covers all points with radius at most 2, so the optimum is 2.

Suppose a polynomial algorithm had approximation factor

$$2 - \varepsilon$$

for some fixed $\varepsilon > 0$. On an instance of optimum 1, it would return a solution of radius strictly smaller than 2. Since all possible radii are $(0,1,2)$, its radius would be at most 1. On an instance of optimum 2, every feasible solution has radius 2. The algorithm would therefore decide VERTEX COVER in polynomial time.

Thus:

Theorem 13 Unless $P = NP$, no polynomial-time algorithm for arbitrary finite metrics can guarantee an approximation factor strictly smaller than 2.

Hence the farthest-first approximation guarantee is best possible in the standard worst-case complexity sense.

8.4 15. Boundary and sanity checks

$k = n$ Taking $F = X$ gives

$$\delta(X, X) = 0.$$

The dynamic program returns the partition into singleton blocks.

$k = 1$ **on the line** Only the two extreme points matter. An optimal center is a point of X nearest to

$$\frac{x_1 + x_n}{2},$$

and the optimum is

$$\min_{1 \leq \ell \leq n} \max x_\ell - x_1, x_n - x_\ell.$$

Nonuniqueness For

$$X = 0, 2, \quad k = 1,$$

both $(\{0\})$ and $(\{2\})$ are optimal, with radius 2.

A two-cluster example For

$$X = 0, 1, 10, 11, \quad k = 2,$$

the optimal consecutive partition is

$$0, 1 \mid 10, 11.$$

Each block has discrete one-center radius 1, so an optimal answer is

$$F = 1, 11, \quad \delta(X, F) = 1.$$

Why the order is decisive On the line, nearest-center cells are consecutive intervals, producing the dynamic program. In an arbitrary metric, the radius- r neighborhoods can encode essentially arbitrary graph domination instances, which is exactly what causes NP-completeness.

8.5 Final answer

For **Q764a**, the problem admits an exact quadratic-time solution. Compute the optimal one-center radius $c(i, j)$ for every consecutive block, then use the bottleneck recurrence

$$\rho_p(j) = \min_{1 \leq i \leq j} \max \rho_{p-1}(i-1), c(i, j).$$

Monotonicity reduces each dynamic-programming layer to linear time, giving total complexity

$$O(n^2 + kn) = O(n^2).$$

For **Q764b**, the exact problem is NP-complete even for metrics with distances only 1 and 2. An exact solution can be obtained by enumerating all k -subsets in

$$O\left(\binom{n}{k} nk\right)$$

time. In polynomial time, farthest-first selection gives the sharp guarantee

$$\delta(X, F) \leq 2, \delta_{\text{opt}},$$

and no factor below 2 is possible unless $P = NP$.

[1]: <https://www.cs.cornell.edu/courses/cs722/2000sp/karp.pdf> "cs722"

Formal verification. Lean source; online type-check.